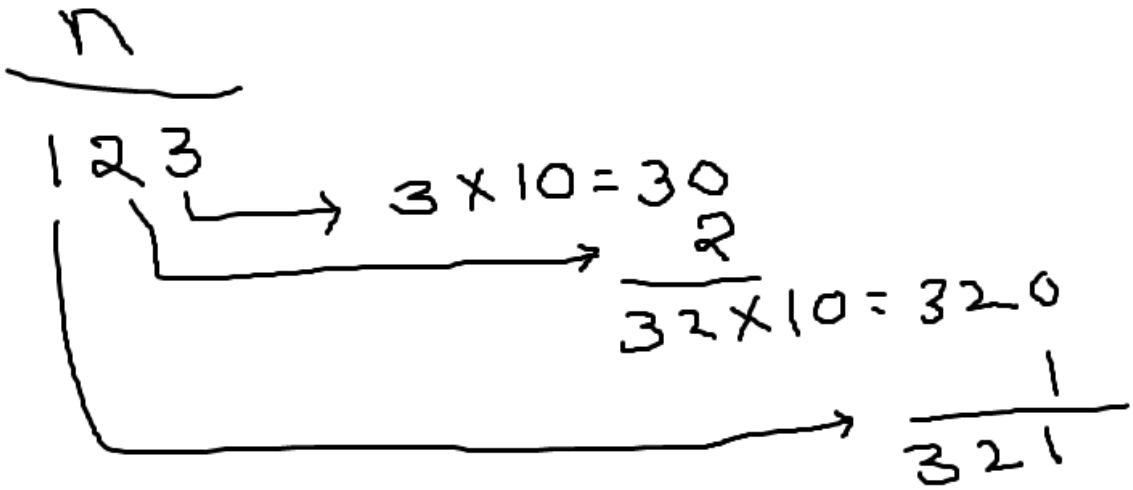
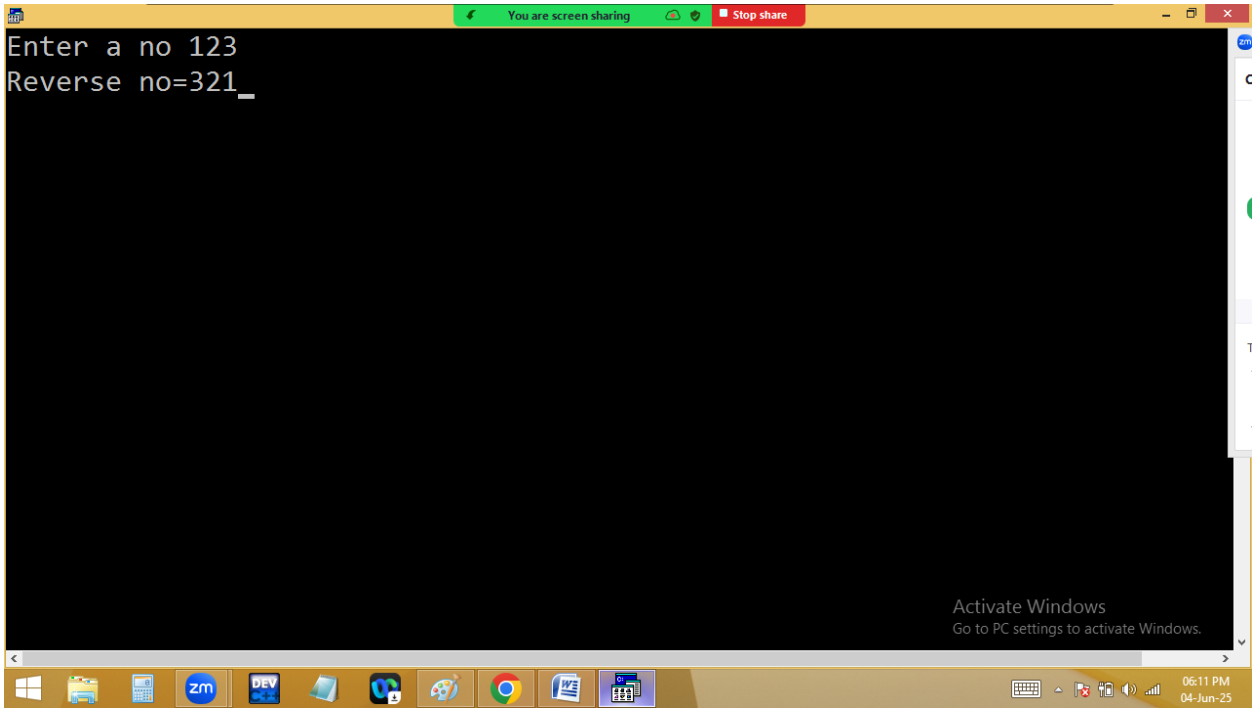
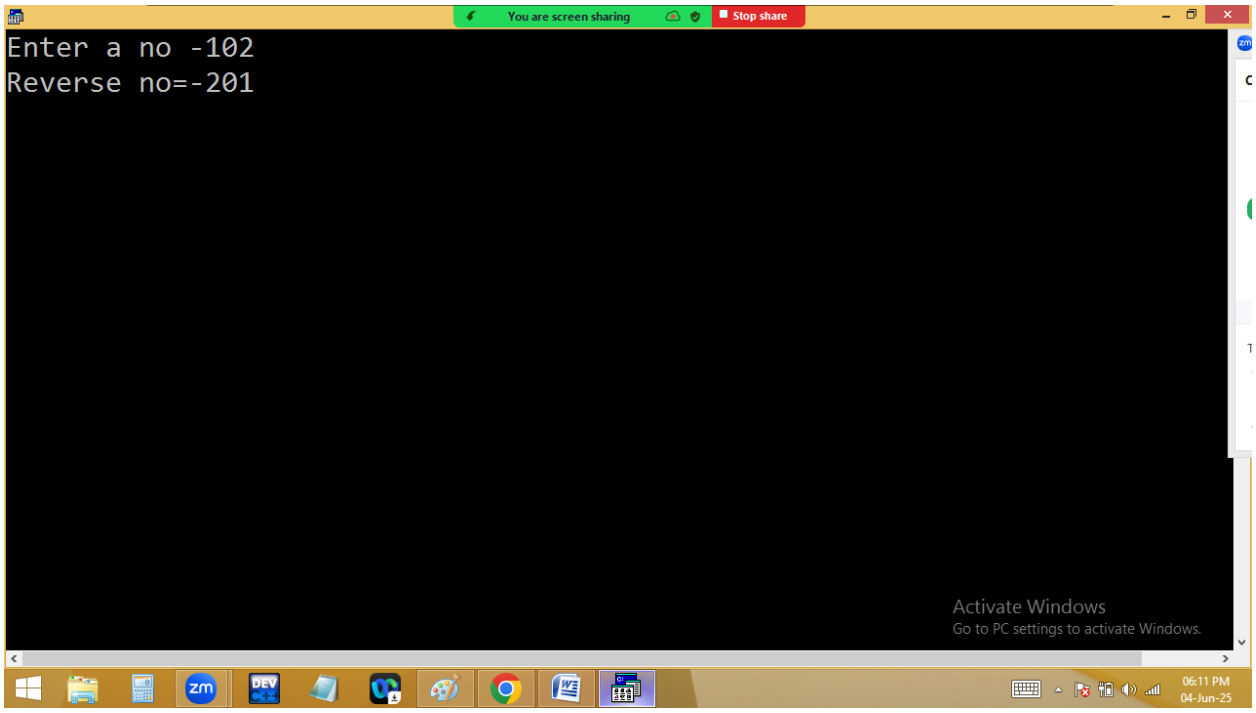
Reverse no?

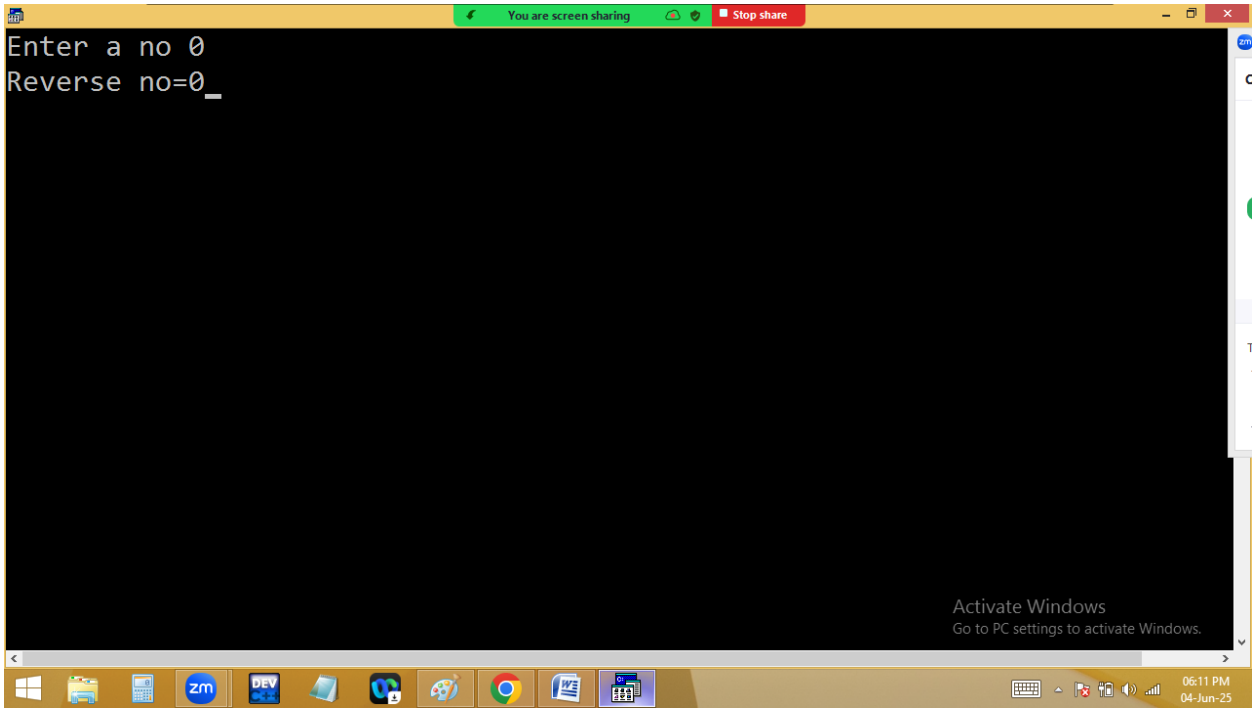
105 ➔ 501

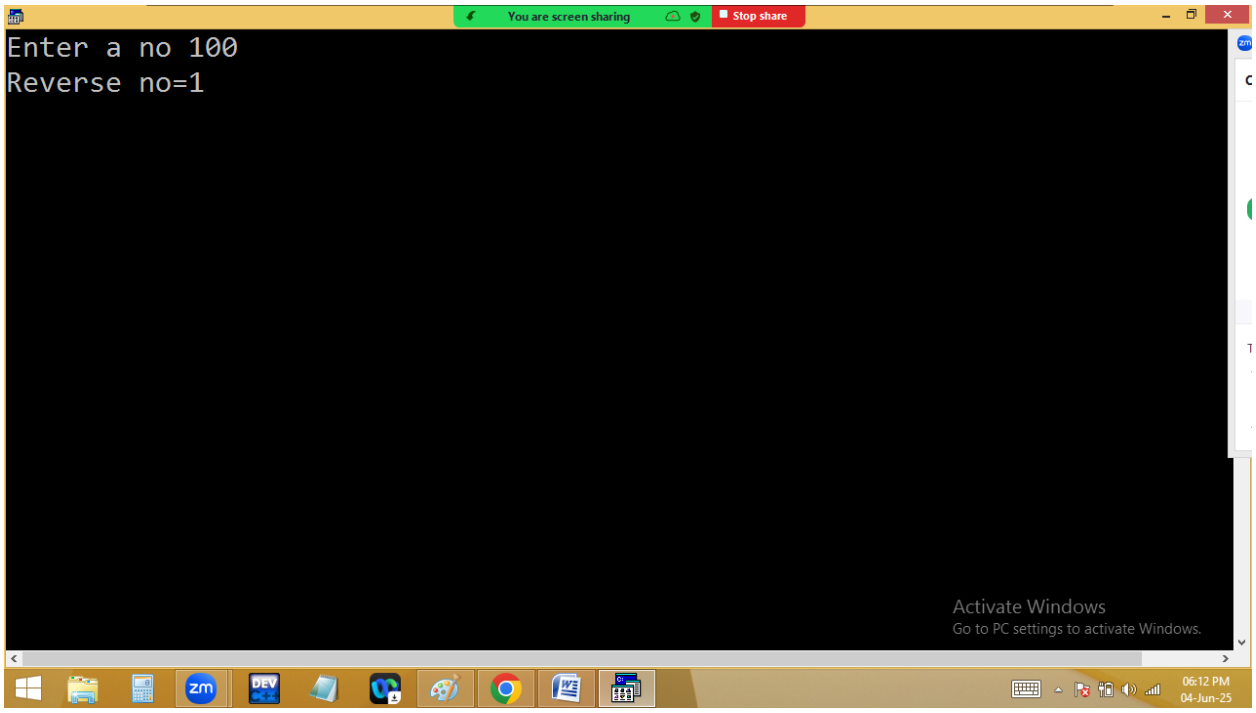


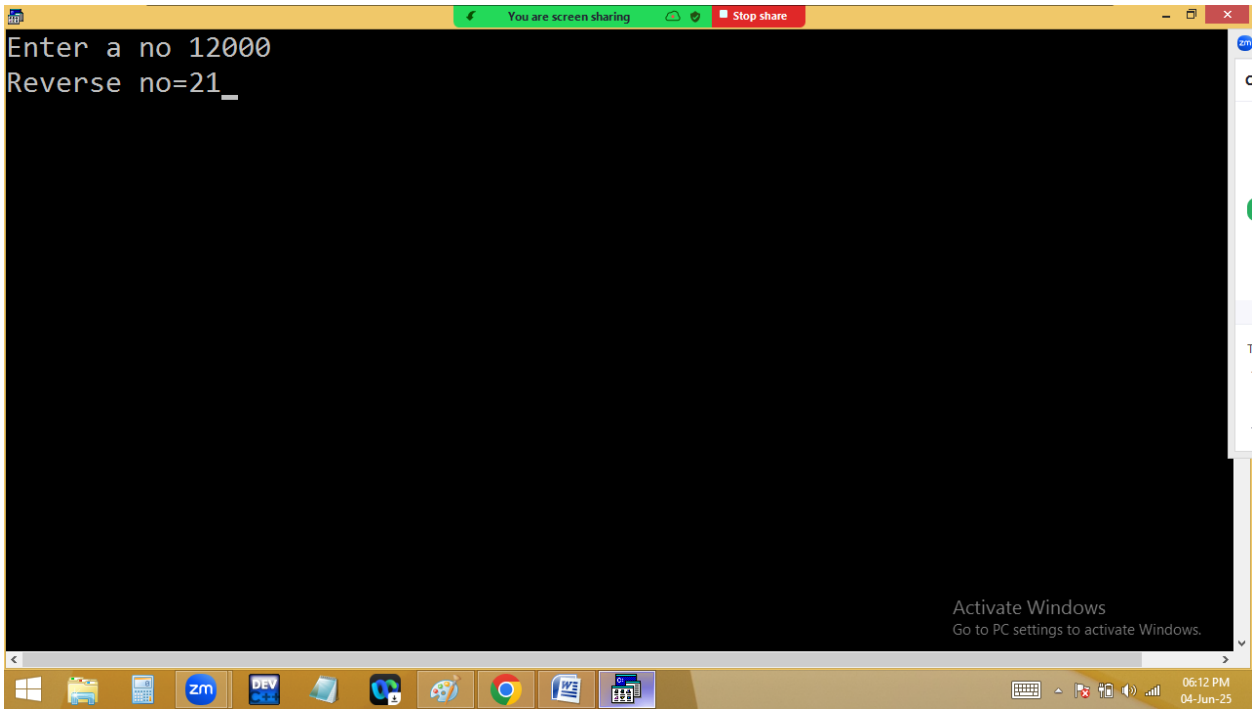

```
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
     Line 13    Col 1   Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long n,rev=0; int r;
clrscr();
printf("Enter a no "); scanf("%ld",&n);
while(n!=0)
{
r=n%10; rev=rev*10+r;n=n/10;
}
printf("Reverse no=%ld",rev);
getch();
}
```

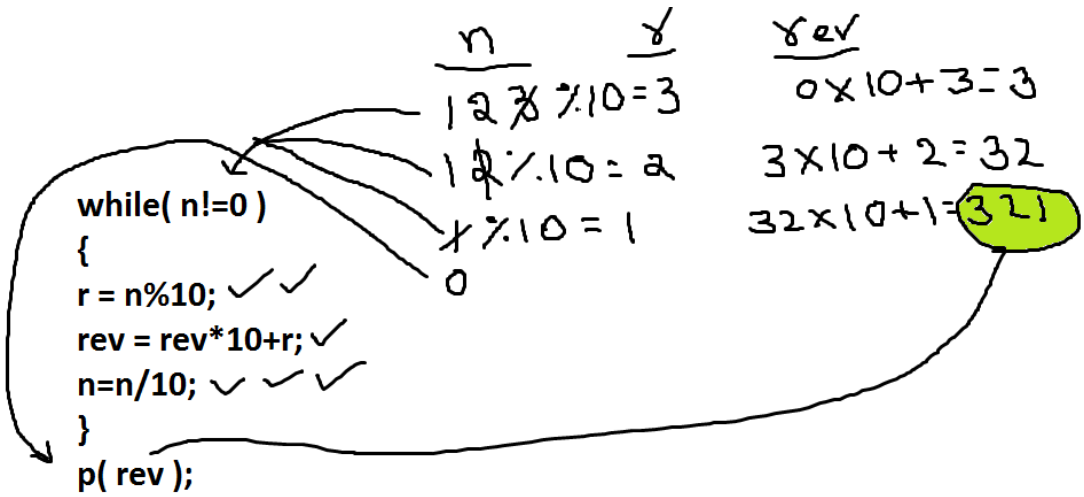

You are screen sharing
Stop share
Enter a no 123
Reverse no=321
Activate Windows
Go to PC settings to activate Windows.
06:11 PM
04-Jun-25

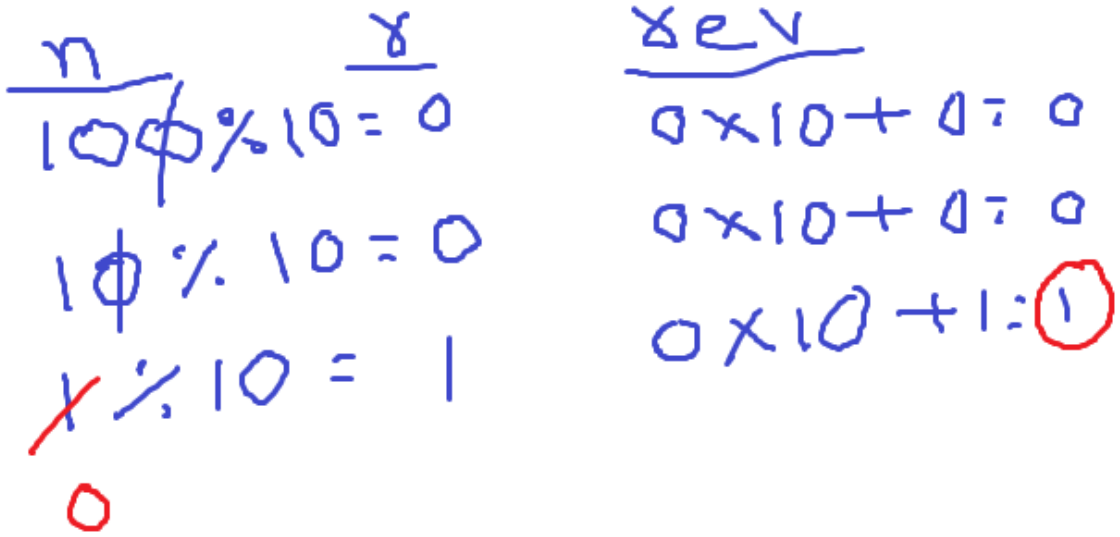

You are screen sharing
Stop share
Enter a no -102
Reverse no=-201
Activate Windows
Go to PC settings to activate Windows.
06:11 PM
04-Jun-25


You are screen sharing
Stop share
Enter a no 0
Reverse no=0
Activate Windows
Go to PC settings to activate Windows.
06:11 PM
04-Jun-25


You are screen sharing
Stop share
Enter a no 100
Reverse no=1
Activate Windows
Go to PC settings to activate Windows.
06:12 PM
04-Jun-25

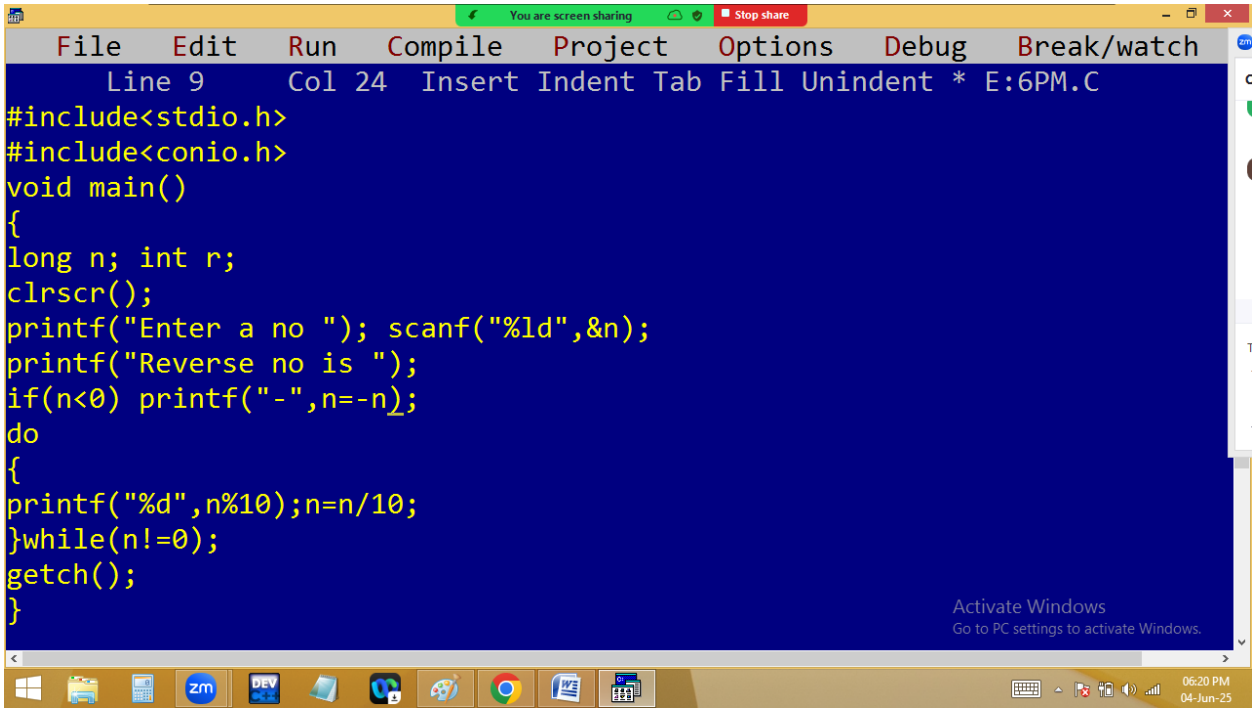

You are screen sharing
Stop share
Enter a no 12000
Reverse no=21
Activate Windows
Go to PC settings to activate Windows.
06:12 PM
04-Jun-25

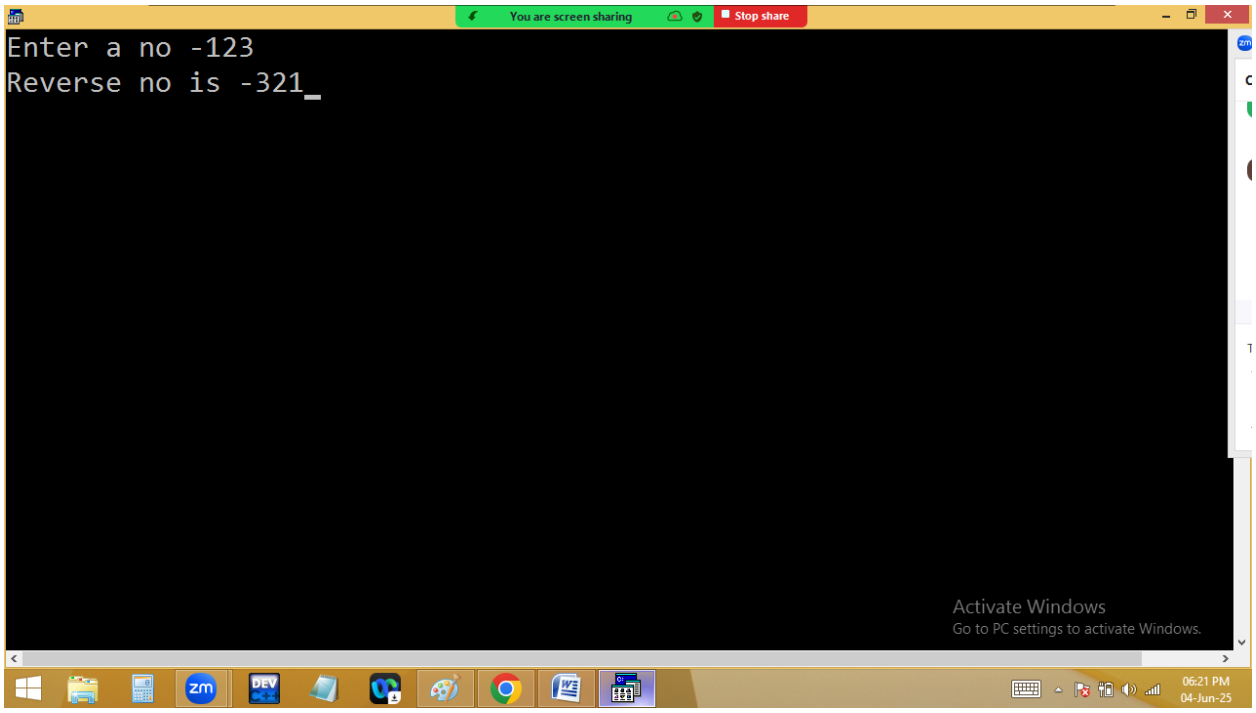

while(n!=0)
{
r = n%10;
rev = rev*10+r;
n=n/10;
}
p(rev);
n
r
12 %10=3
1 %10=2
%10=1
0
rev
0x10+3=3
3x10+2=32
32x10+1=321

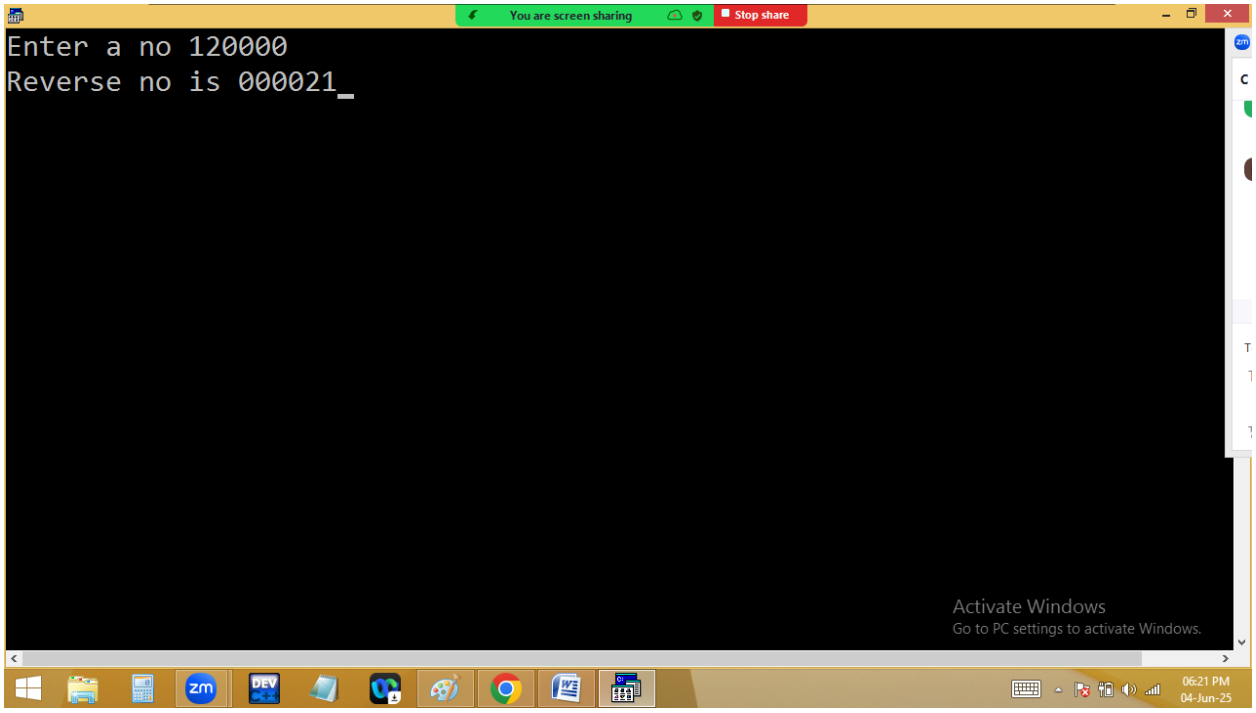


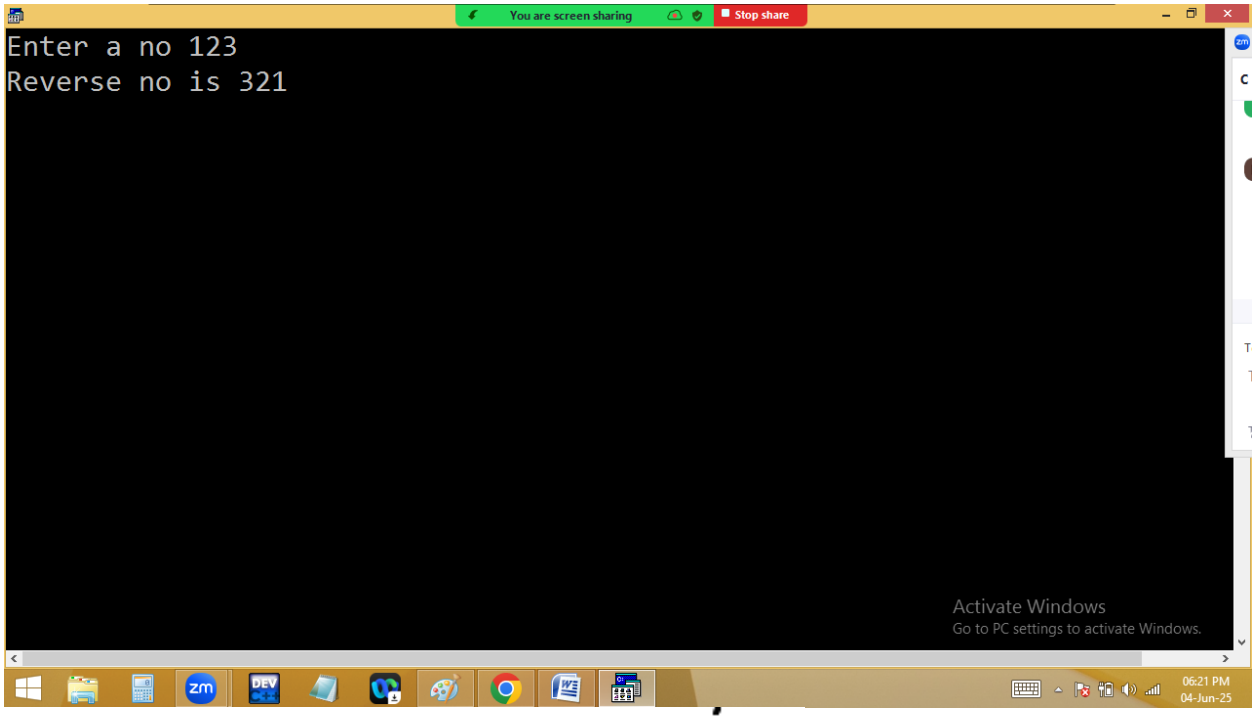
method 2:

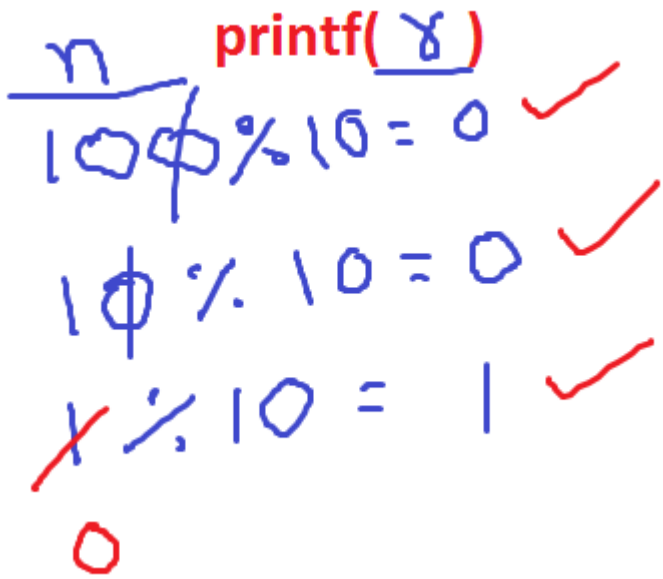
Printing 100 as 001


You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 24 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long n; int r;
clrscr();
printf("Enter a no "); scanf("%ld",&n);
printf("Reverse no is ");
if(n<0) printf("-",n=-n);
do
{
printf("%d",n%10);n=n/10;
}while(n!=0);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:20 PM
04-Jun-25


You are screen sharing
Stop share
Enter a no -123
Reverse no is -321
Activate Windows
Go to PC settings to activate Windows.
06:21 PM
04-Jun-25

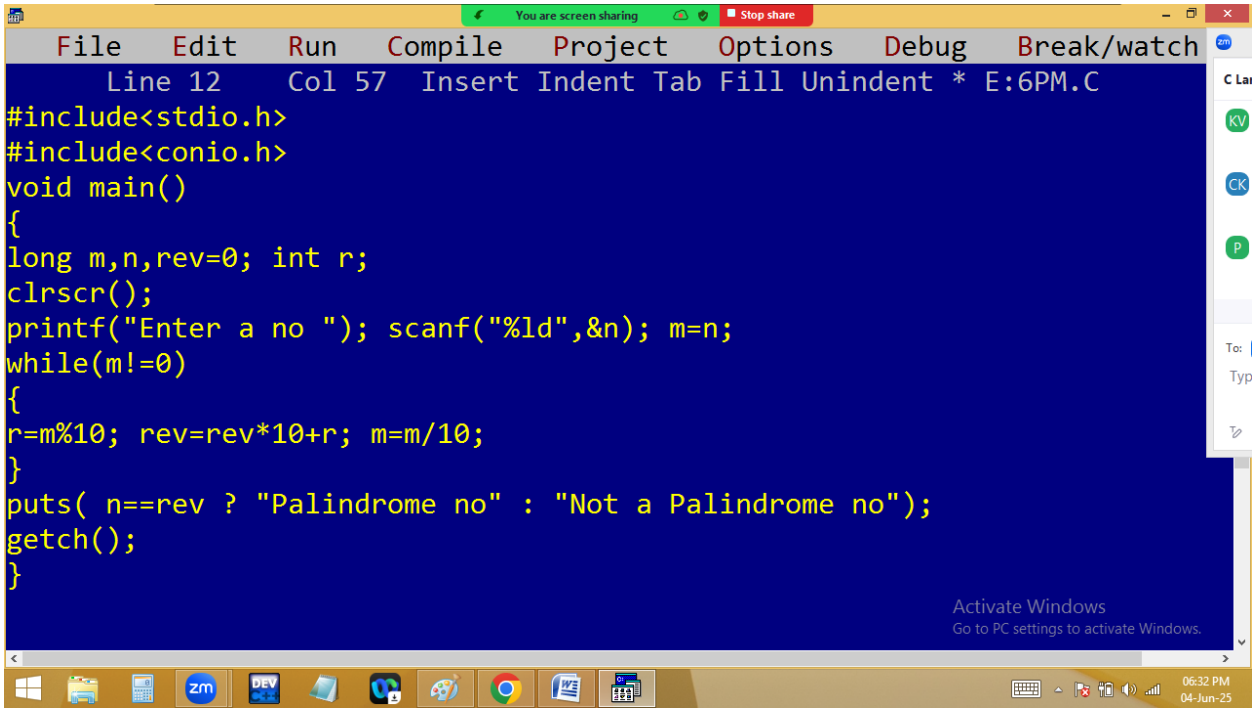

You are screen sharing
Stop share
Enter a no 120000
Reverse no is 000021
Activate Windows
Go to PC settings to activate Windows.
06:21 PM
04-Jun-25

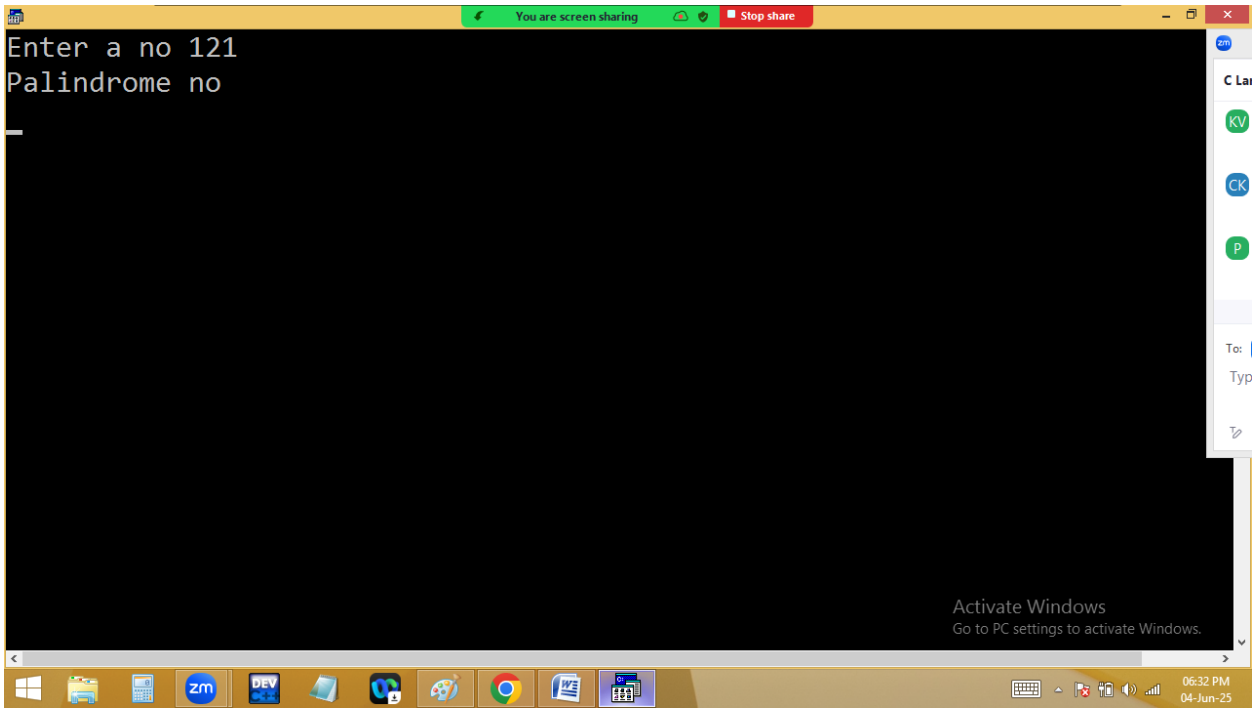


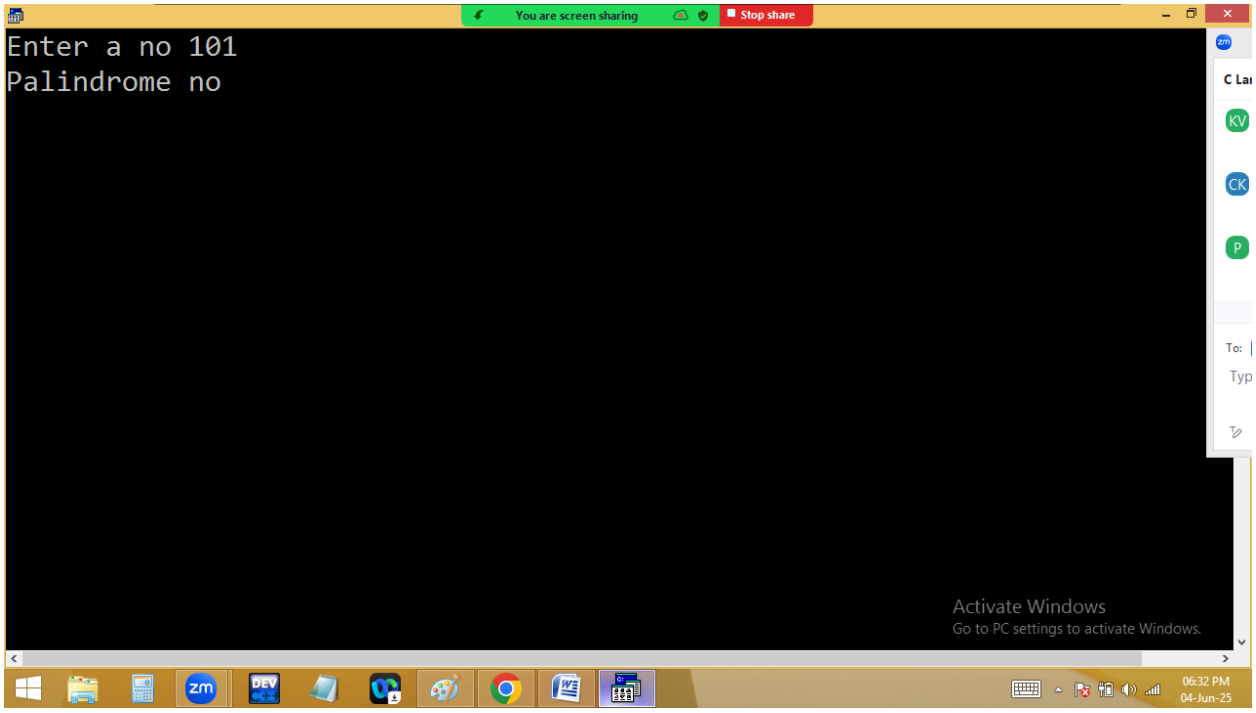


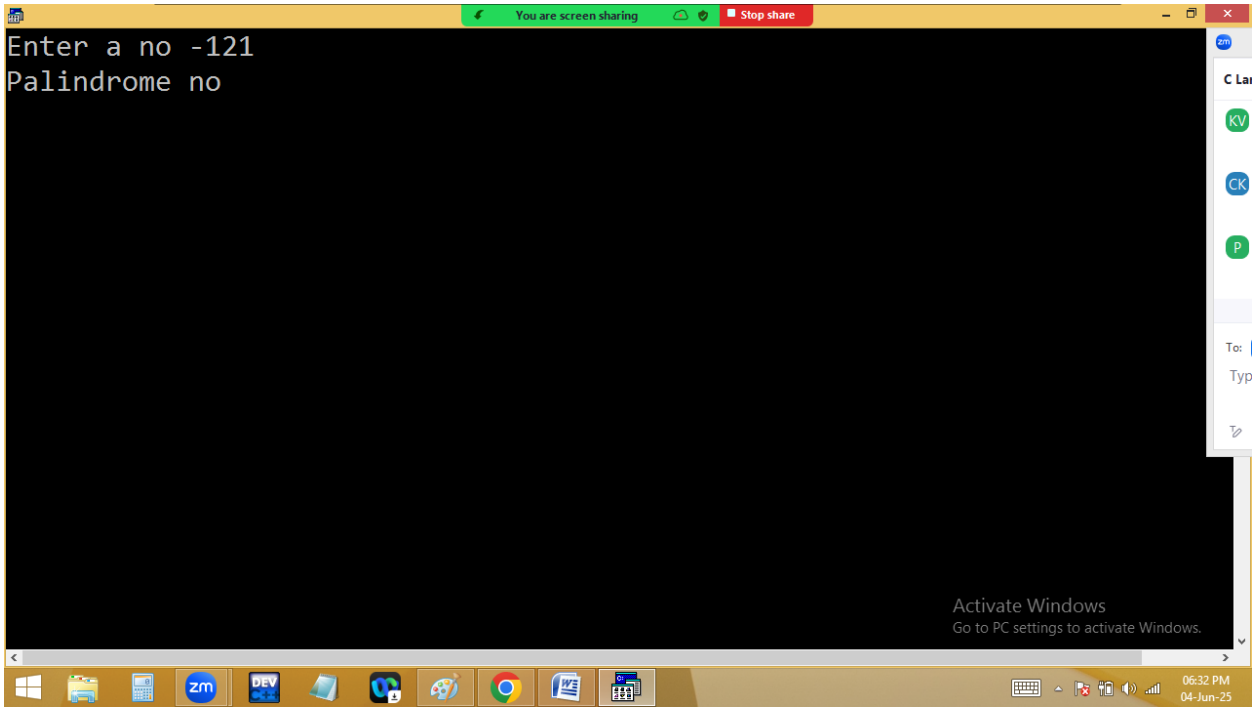
Palindrome no? given no and reverse no both are same.

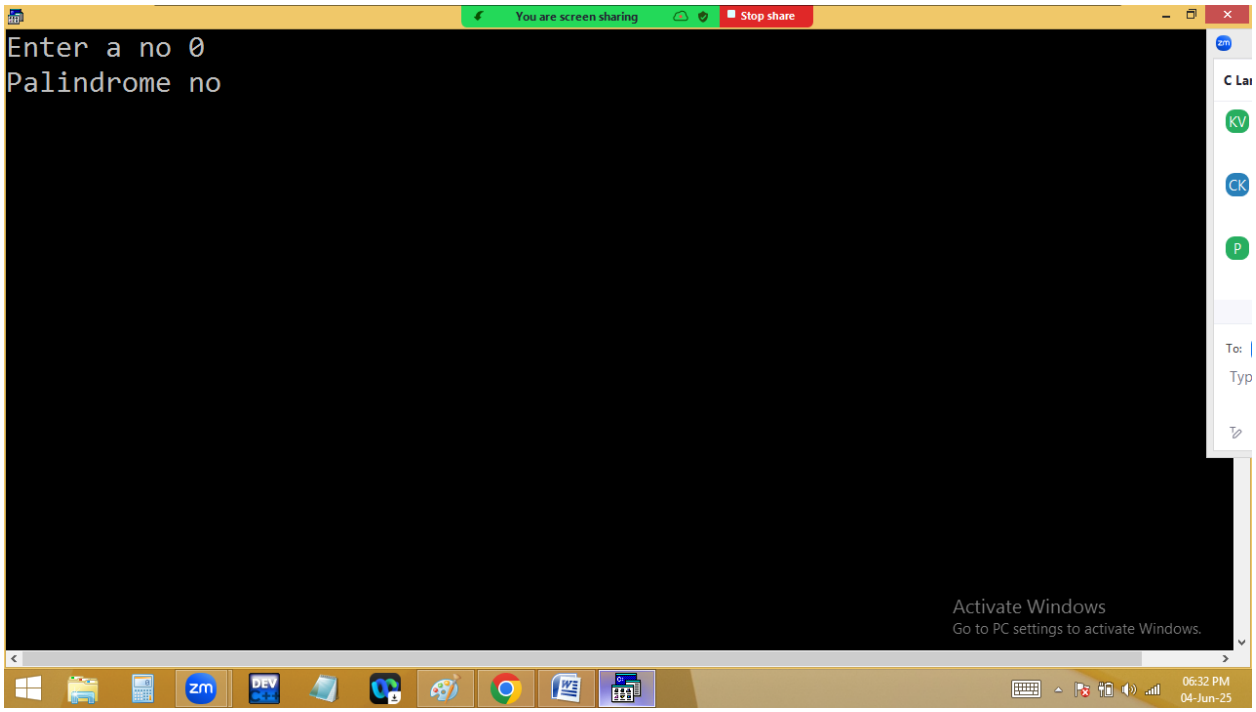
101 reverse also 101 ← palindrome no

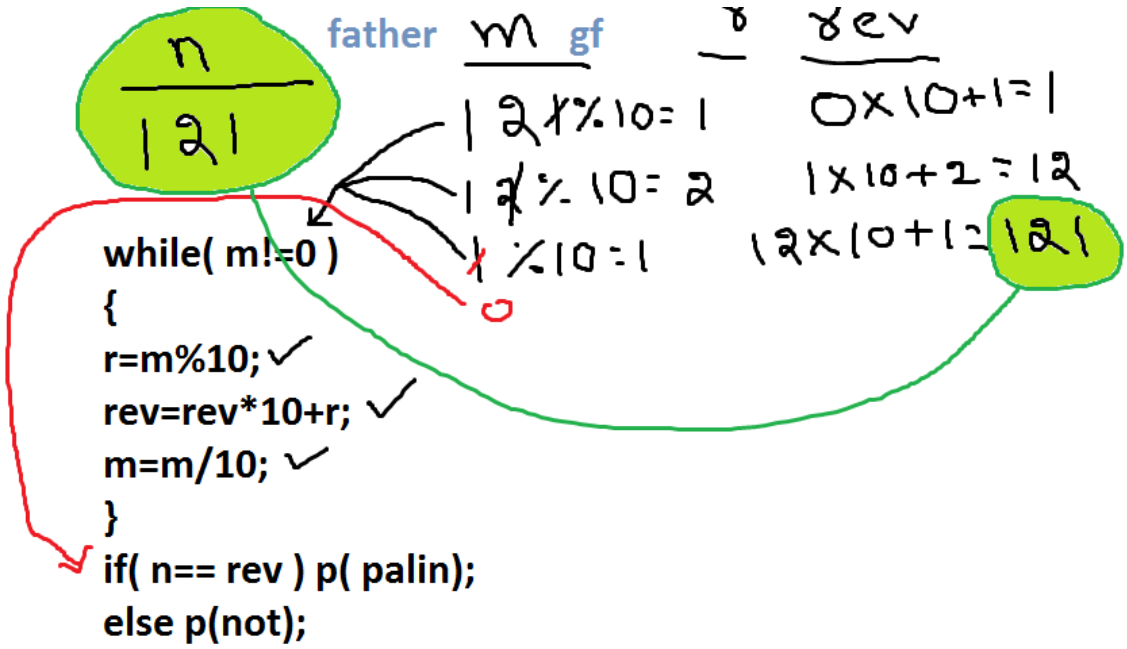

You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 57 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long m,n,rev=0; int r;
clrscr();
printf("Enter a no "); scanf("%ld",&n); m=n;
while(m!=0)
{
r=m%10; rev=rev*10+r; m=m/10;
}
puts(n==rev ? "Palindrome no" : "Not a Palindrome no");
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:32 PM
04-Jun-25


You are screen sharing
Stop share
Enter a no 121
Palindrome no
Activate Windows
Go to PC settings to activate Windows.
06:32 PM
04-Jun-25


You are screen sharing
Stop share
Enter a no 101
Palindrome no
Activate Windows
Go to PC settings to activate Windows.
06:32 PM
04-Jun-25

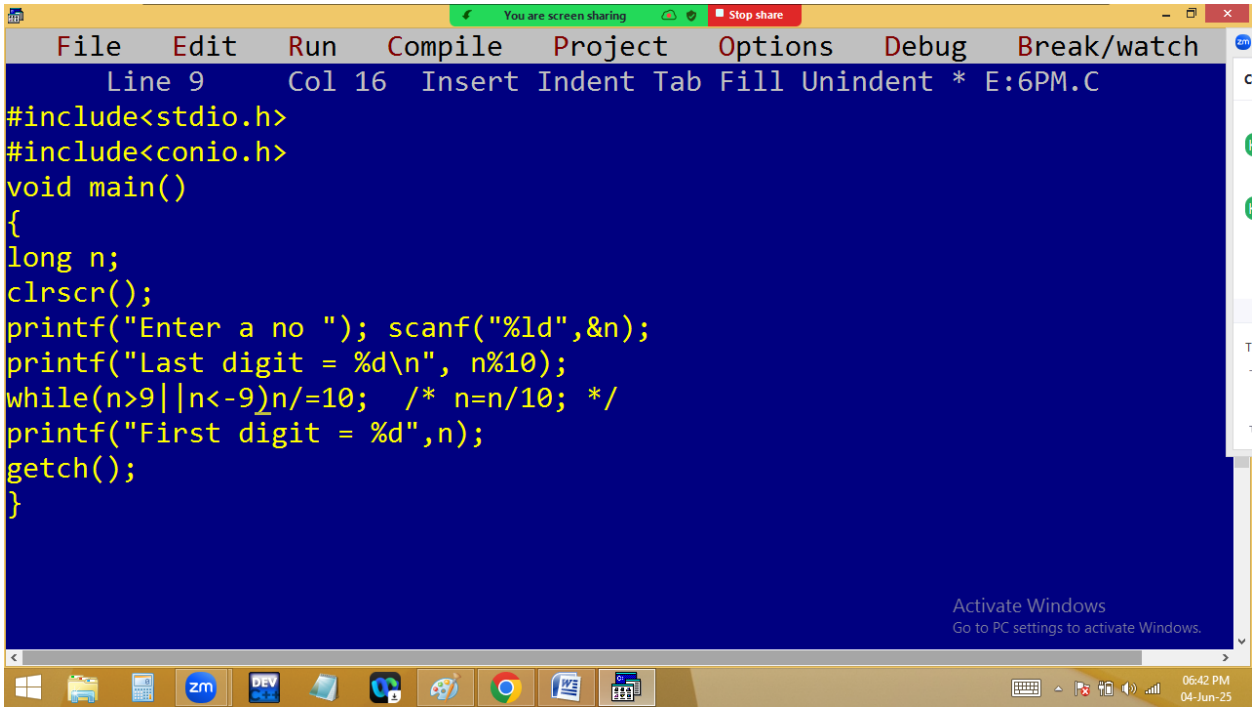

You are screen sharing
Stop share
Enter a no -121
Palindrome no
Activate Windows
Go to PC settings to activate Windows.
06:32 PM
04-Jun-25

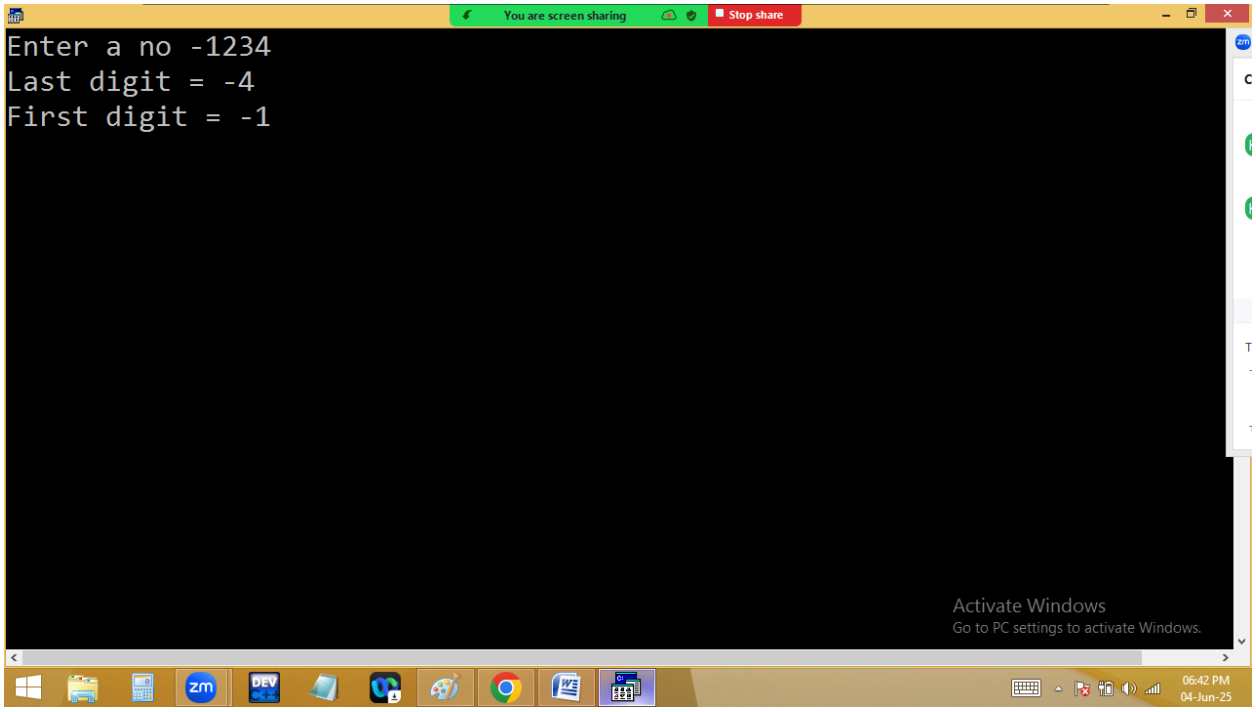


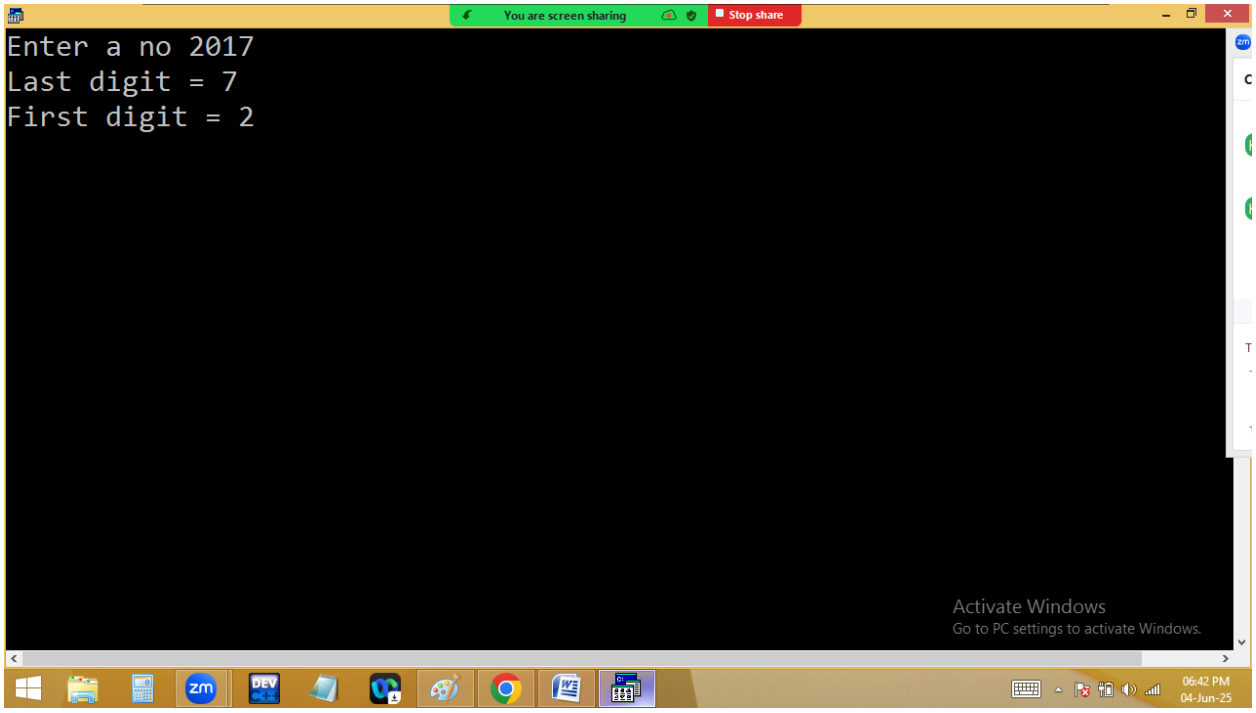


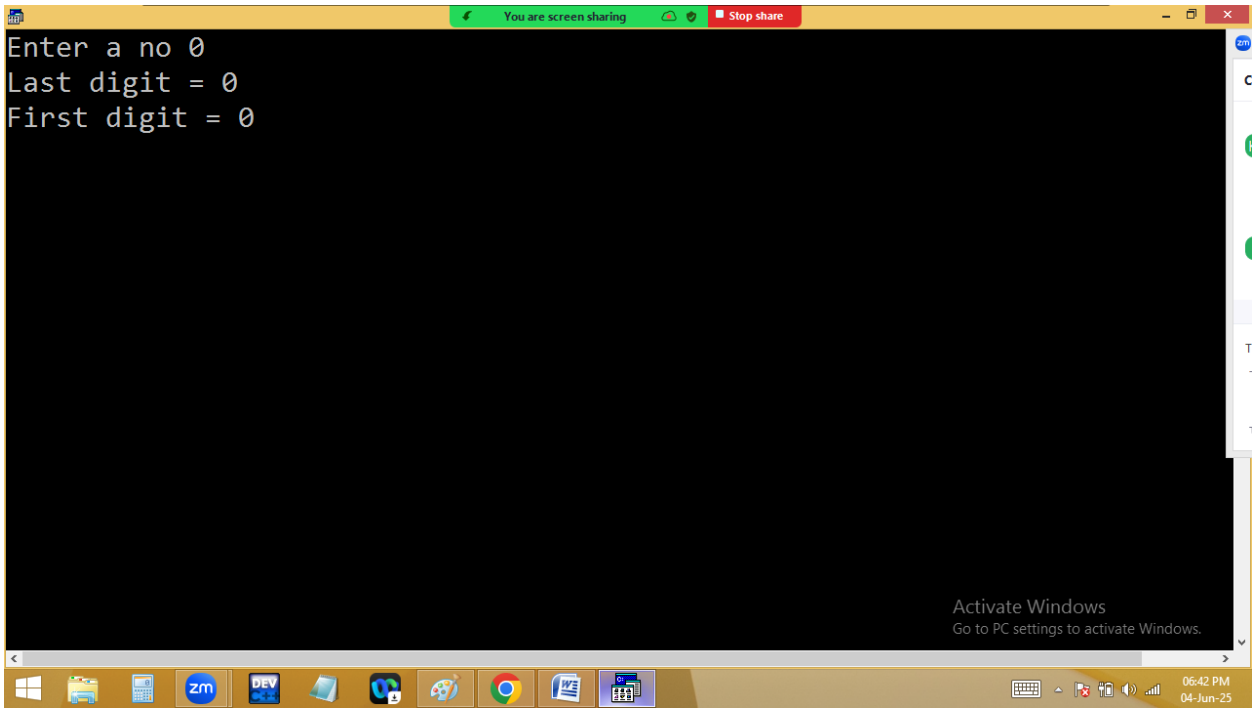
Finding first and last digits of given no?

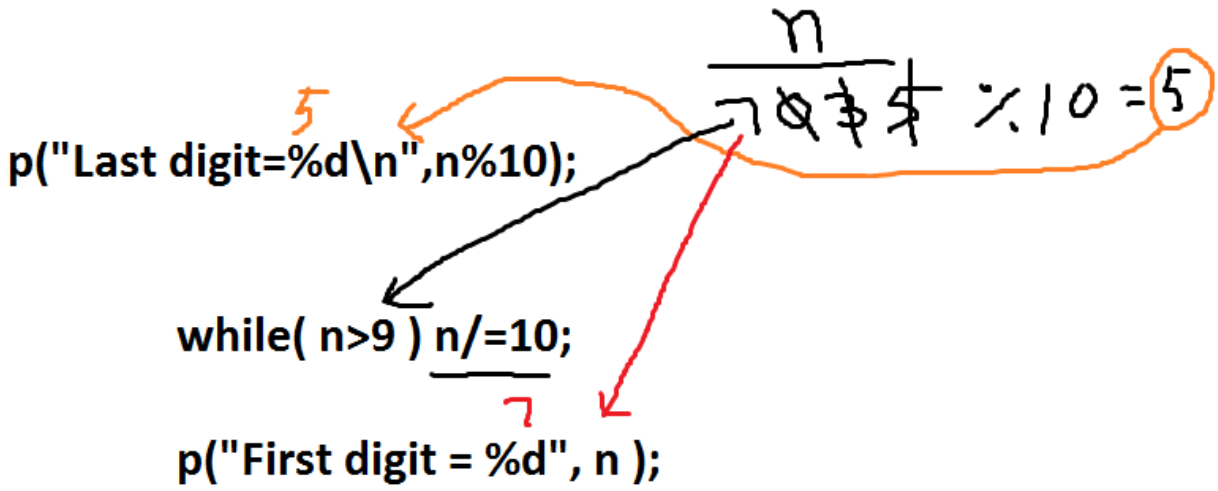
Eg: 7135 ➔ 5 is last digit and 7 is first digit






You are screen sharing
Stop share
Enter a no 2017
Last digit = 7
First digit = 2
Activate Windows
Go to PC settings to activate Windows.
06:42 PM
04-Jun-25


You are screen sharing
Stop share
Enter a no 0
Last digit = 0
First digit = 0
Activate Windows
Go to PC settings to activate Windows.
06:42 PM
04-Jun-25



for loop:

for loop:

It is an entry control loop.

for is a keyword.

It is also used to repeat a program several times based on a condition.

When compared with while and do while, for loop is looking to be smart. In for it is compulsory to maintain two semicolons.

For works without condition also and default condition is always 1 i.e. true.

Generally for loop is having 3 expressions.

1. **Initialization**
2. **Test condition / expression**
3. **Increment/decrement / updation**

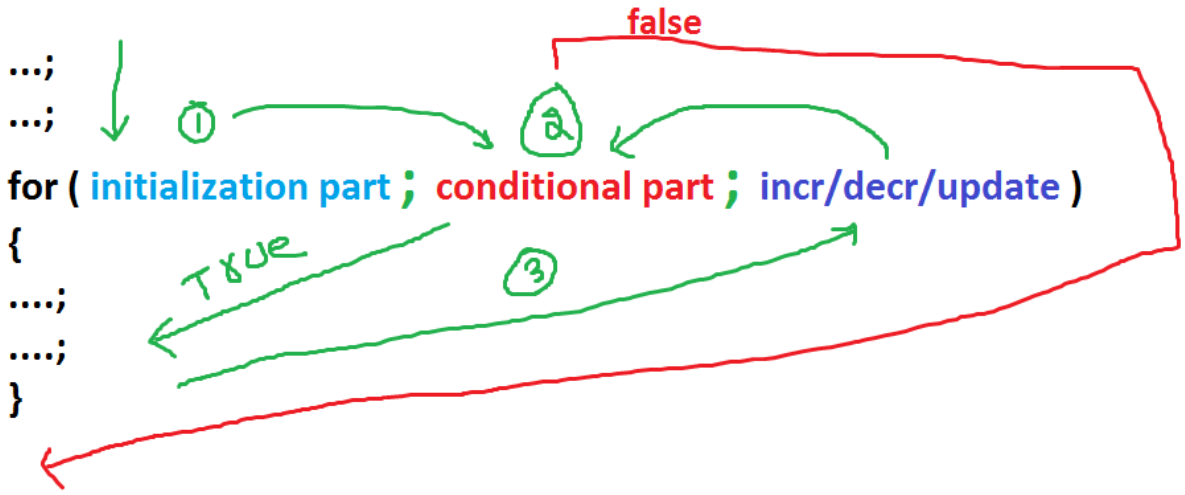
At first entry of for loop the initialization part is executed and later the test condition is checked. If the condition is true then the for block statements are executed. After completion of the block, the increment or decrement part is executed. Later once again the test condition is evaluated. If it is true then once again for block statements are executed. Like this the process is continued until the condition becomes false. Here the

initialization part is executed only once, at the time of loop beginning.

It is mandatory to maintain 2 semicolon (;) in a for loop.

If the for loop is having more than three expressions, it is mandatory to separate the expressions with , separator.

If the for loop is having less than three expressions, then leave the expressions with empty semicolon.



Note: In for loop we must maintain two semicolons ;

```
      1<=5  ;  a++;  a=5
for(   exp ;   exp  ;  exp  )
{
...;
...;
}
```

```
for(  exp,  exp ;  exp, exp ;  exp, exp  )
{
...;
...;
}
```

```
for(   ;    exp    ;  )
{
...;
...;
}
```

```
for(  ;  ;  )
{
...;
...;
}
```

What is the difference b/w while and for?
while never works without condition
for works without condition also because of the default condition is true

Printing given table?

Eg: 9th table

9*1=9

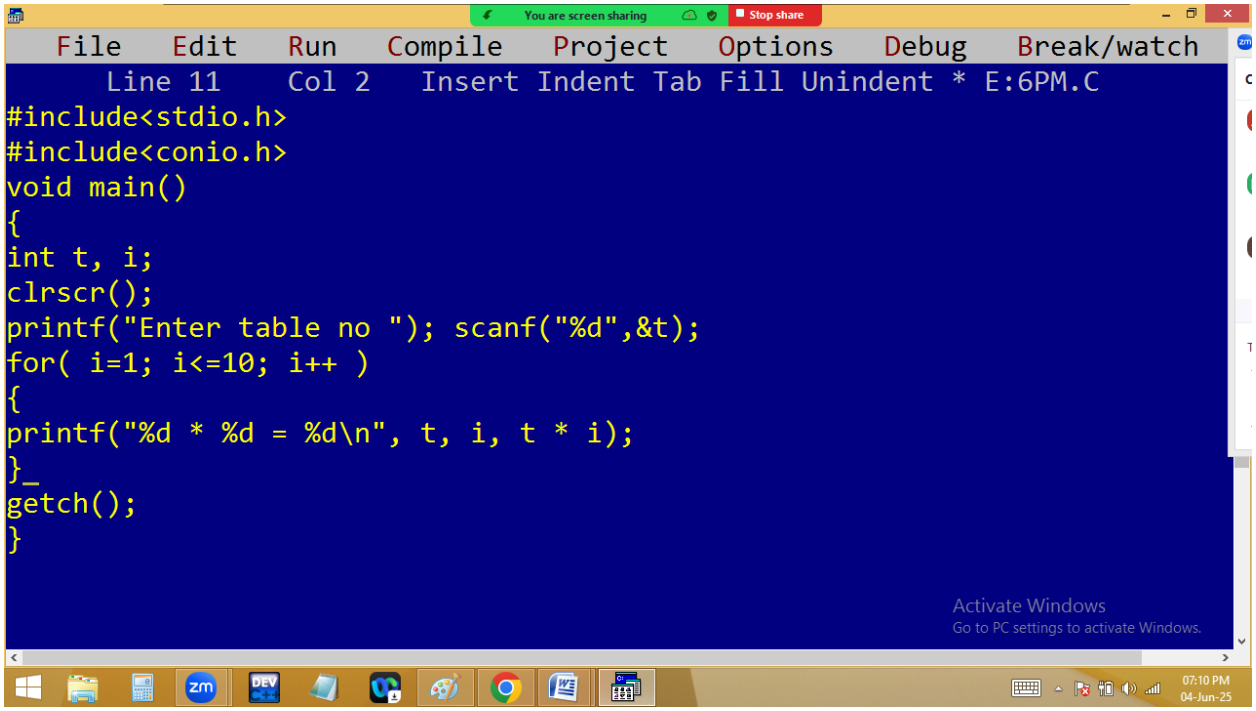
9*2=18

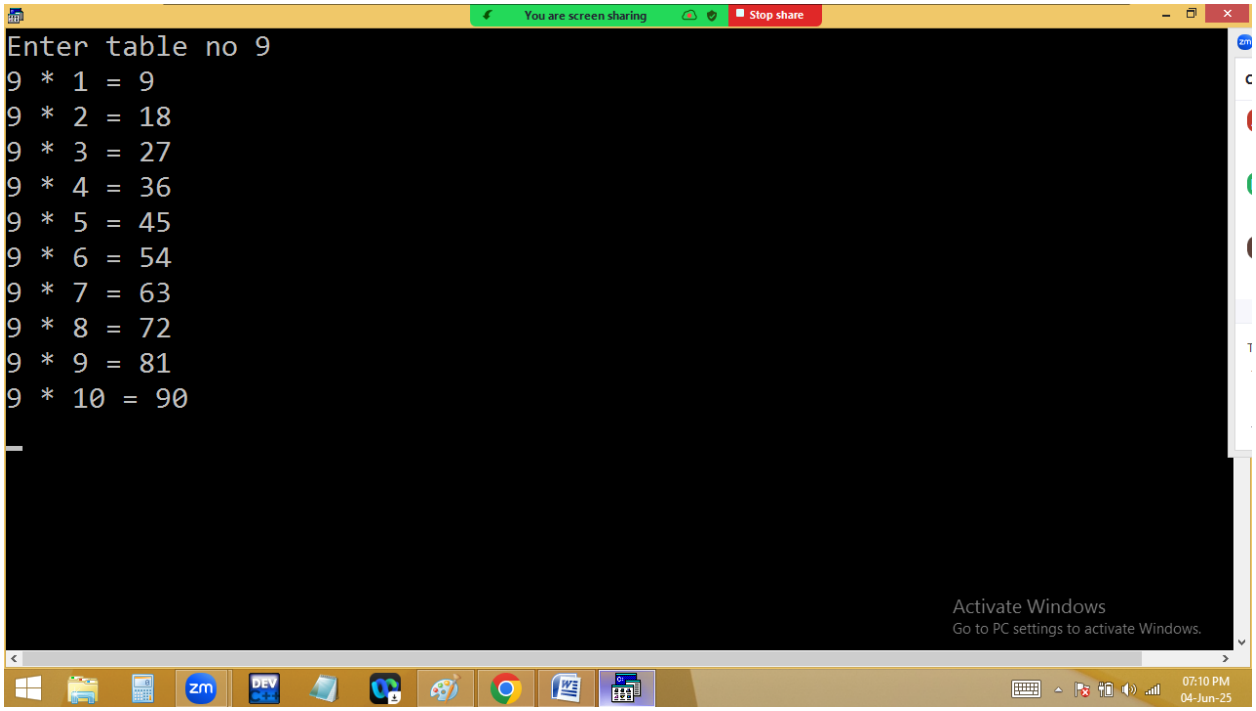
9*3=27

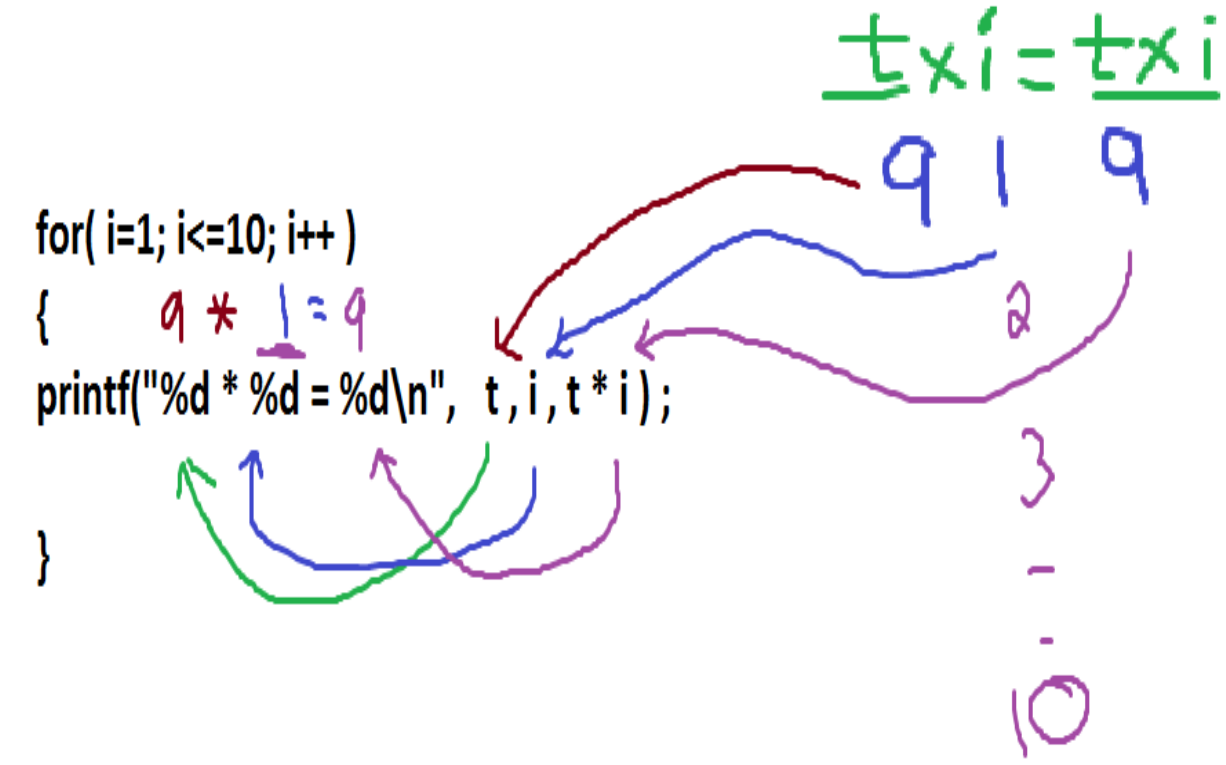
...

...

9*10=90

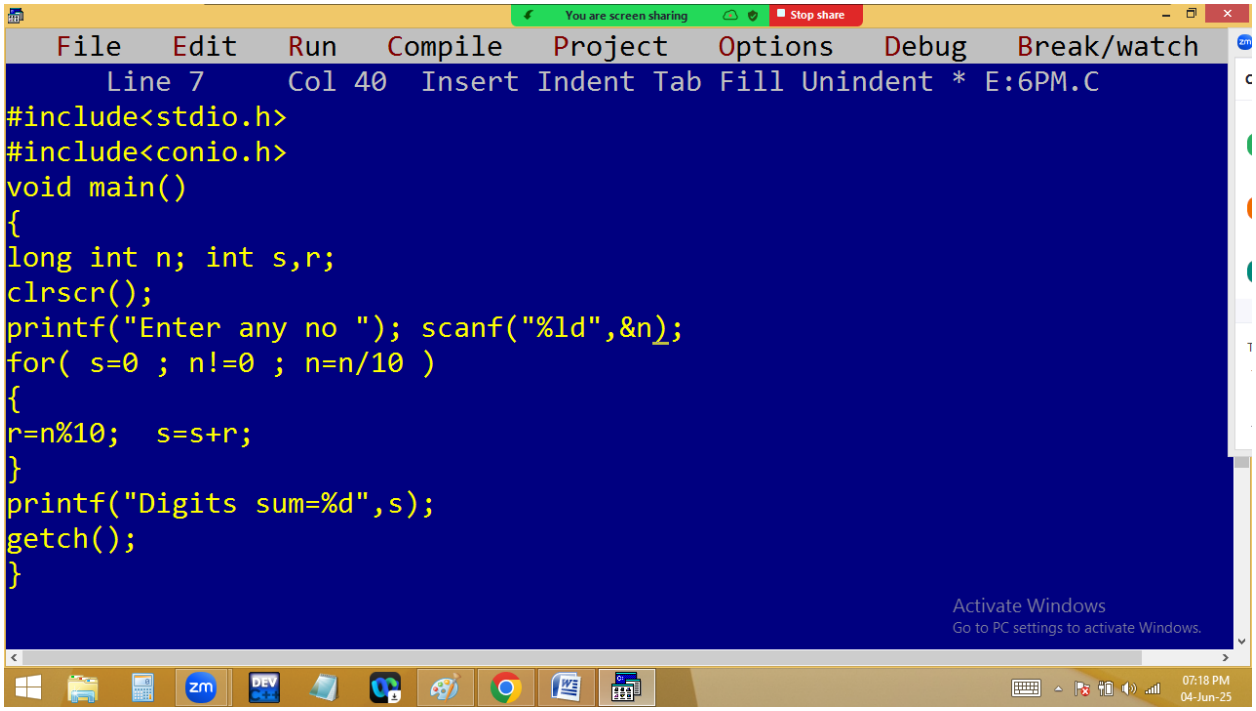

You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 2 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int t, i;
clrscr();
printf("Enter table no "); scanf("%d",&t);
for(i=1; i<=10; i++)
{
printf("%d * %d = %d\n", t, i, t * i);
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:10 PM
04-Jun-25

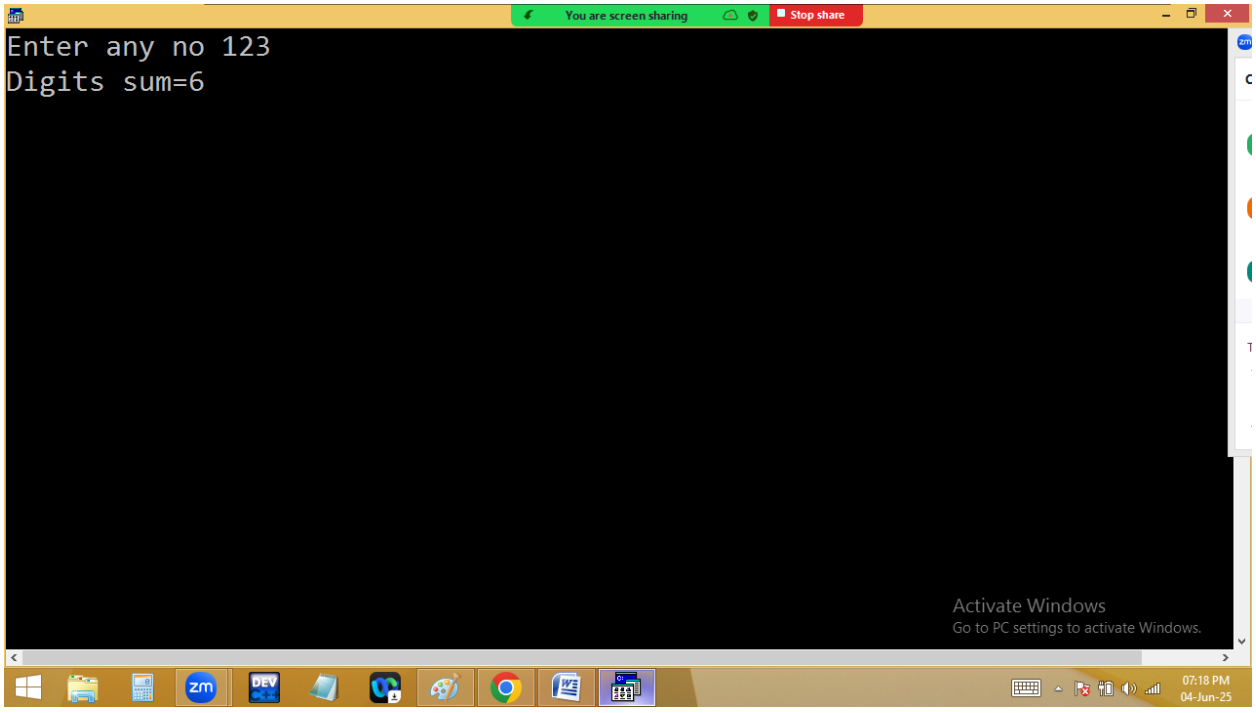

You are screen sharing
Stop share
Enter table no 9
9 * 1 = 9
9 * 2 = 18
9 * 3 = 27
9 * 4 = 36
9 * 5 = 45
9 * 6 = 54
9 * 7 = 63
9 * 8 = 72
9 * 9 = 81
9 * 10 = 90
Activate Windows
Go to PC settings to activate Windows.
07:10 PM
04-Jun-25

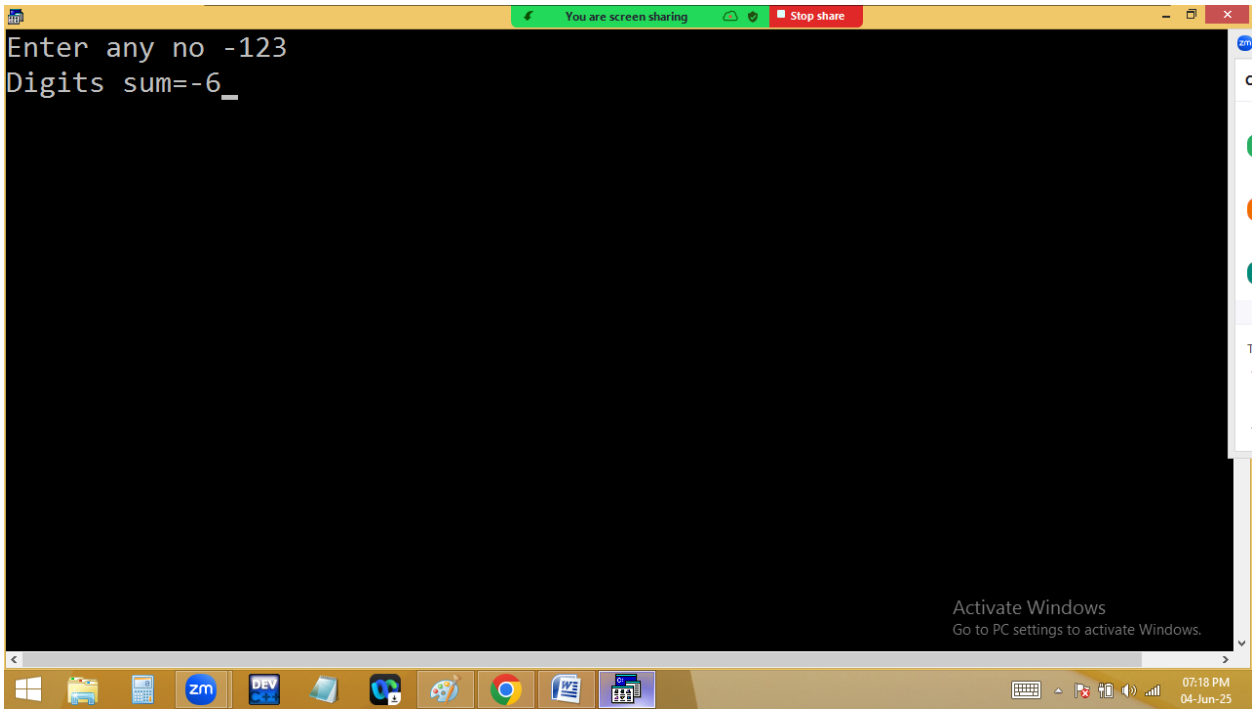

for(i=1; i<=10; i++)
{
printf("%d * %d = %d\n", t, i, t * i);
}
txi = txi
9 1 9
9 * 1 = 9
2
3
-
-
10

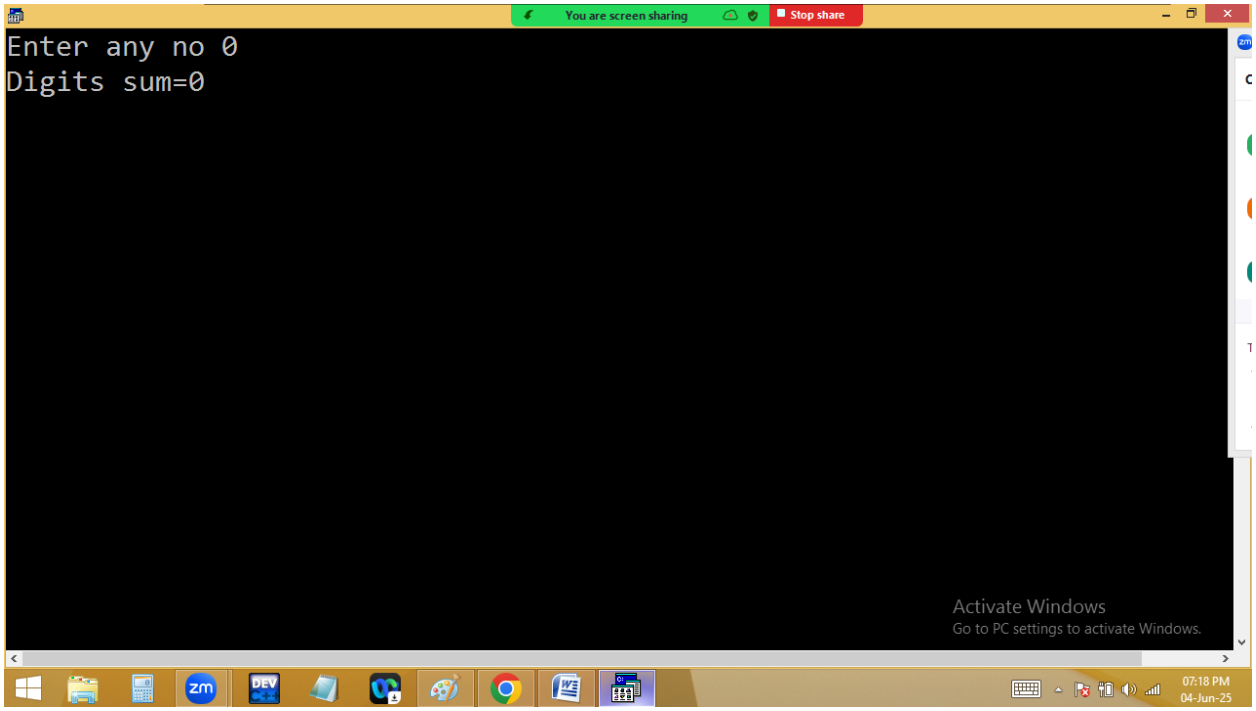
Finding digits sum using for loop?

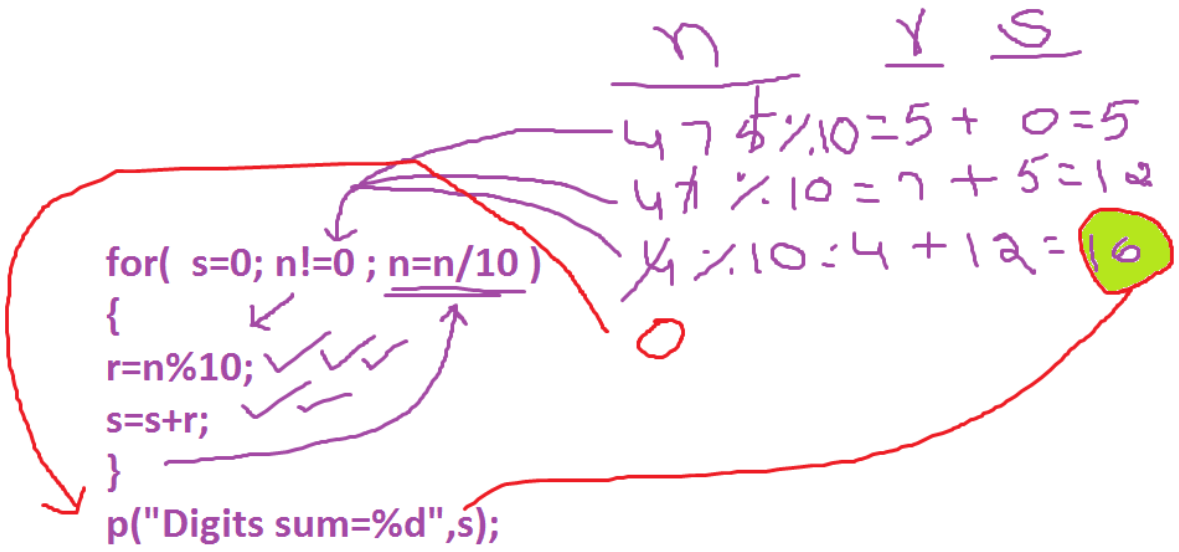
275 digits sum is 2+7+5=14

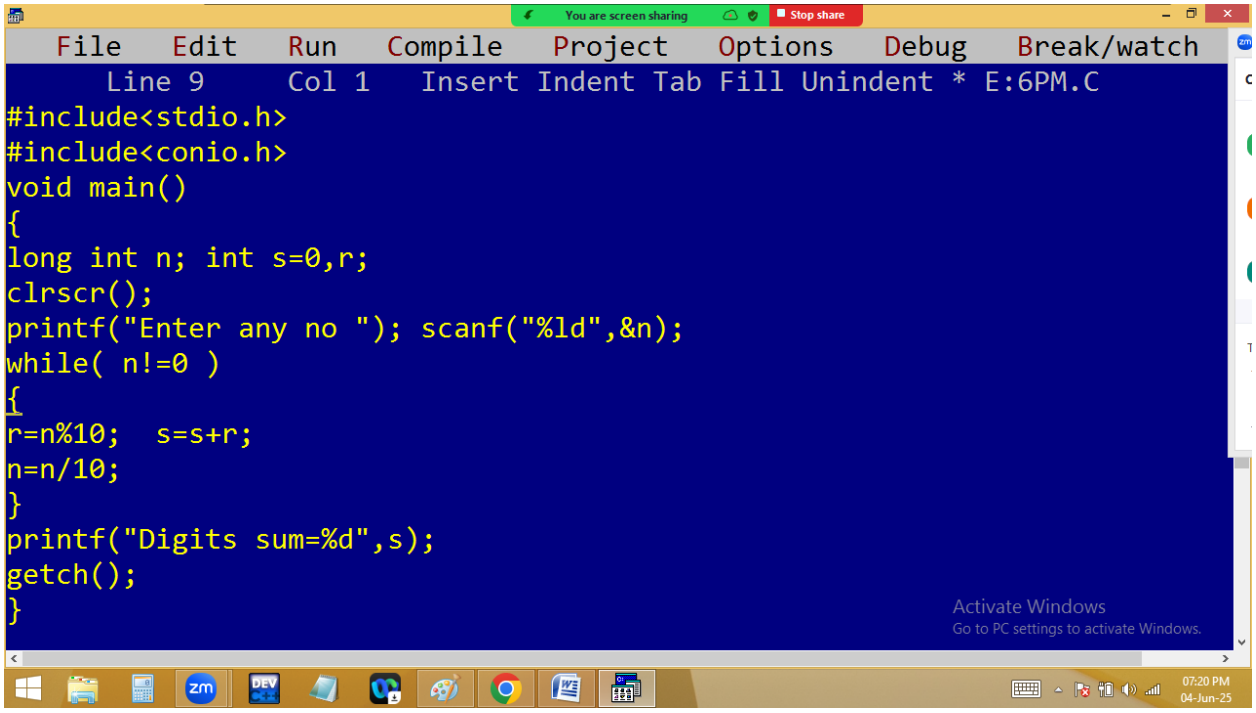


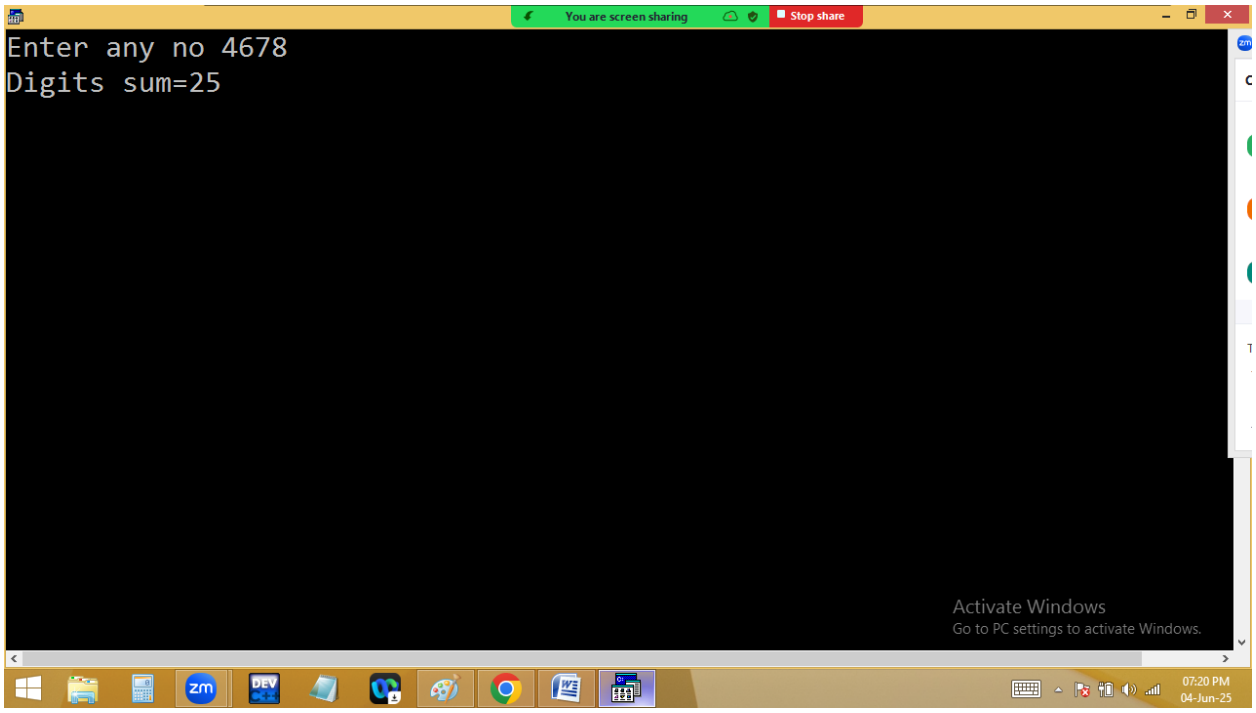


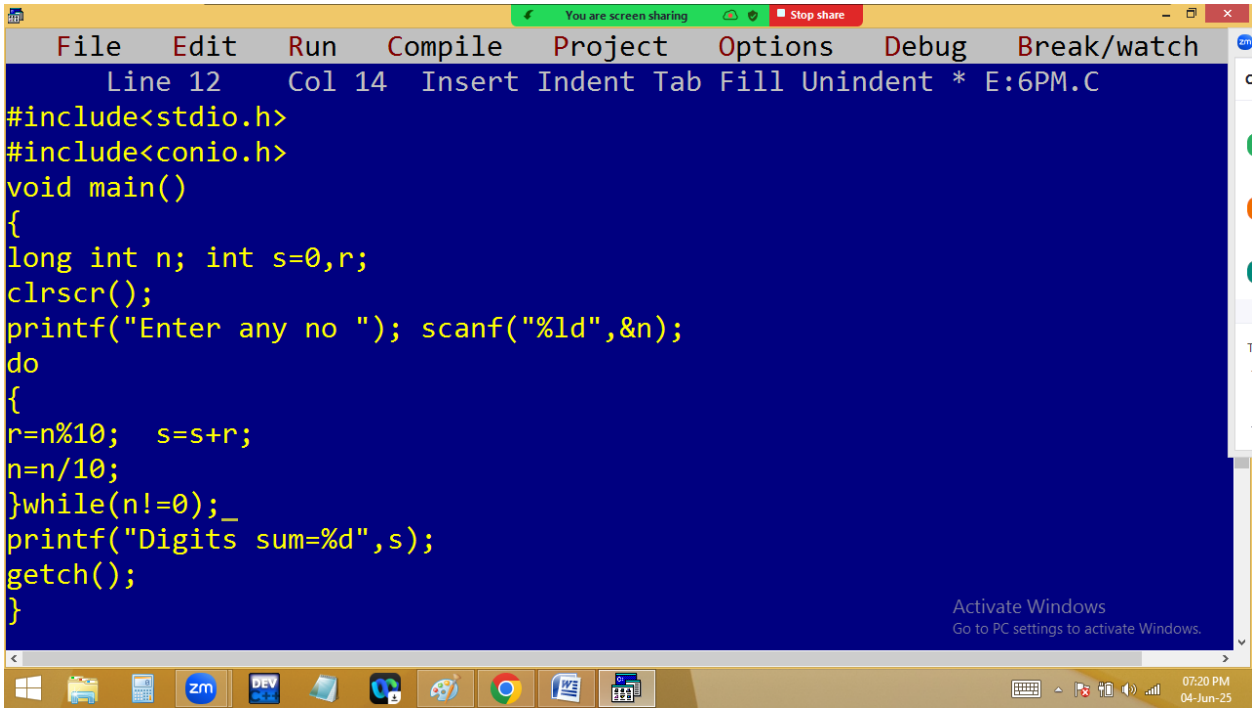

You are screen sharing
Stop share
Enter any no -123
Digits sum=-6
Activate Windows
Go to PC settings to activate Windows.
07:18 PM
04-Jun-25

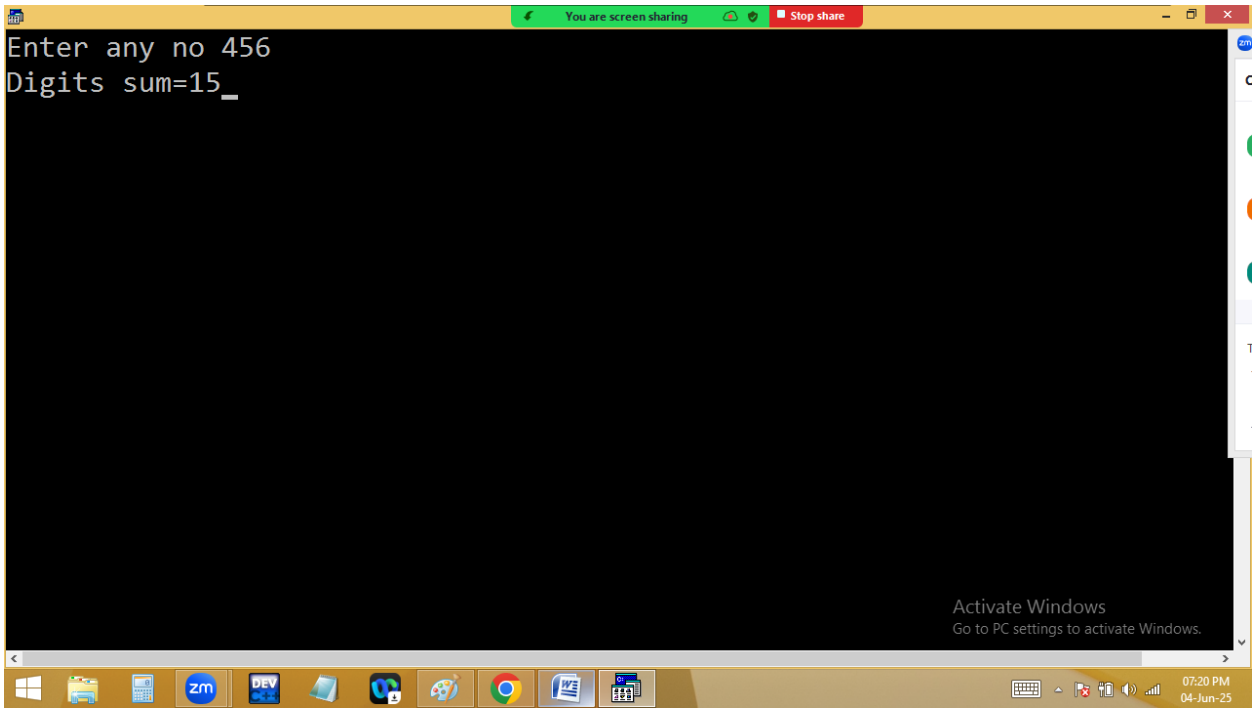

You are screen sharing
Stop share
Enter any no 0
Digits sum=0
Activate Windows
Go to PC settings to activate Windows.
07:18 PM
04-Jun-25


n
r
s
47
47%10=5
s=0+5=5
4%10=4
s=5+7=12
0
4%10=4
s=12+4=16
for(s=0; n!=0 ; n=n/10)
{
r=n%10;
s=s+r;
}
p("Digits sum=%d",s);


You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 1 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long int n; int s=0,r;
clrscr();
printf("Enter any no "); scanf("%ld",&n);
while(n!=0)
{
r=n%10; s=s+r;
n=n/10;
}
printf("Digits sum=%d",s);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:20 PM
04-Jun-25


You are screen sharing
Stop share
Enter any no 4678
Digits sum=25
Activate Windows
Go to PC settings to activate Windows.
07:20 PM
04-Jun-25

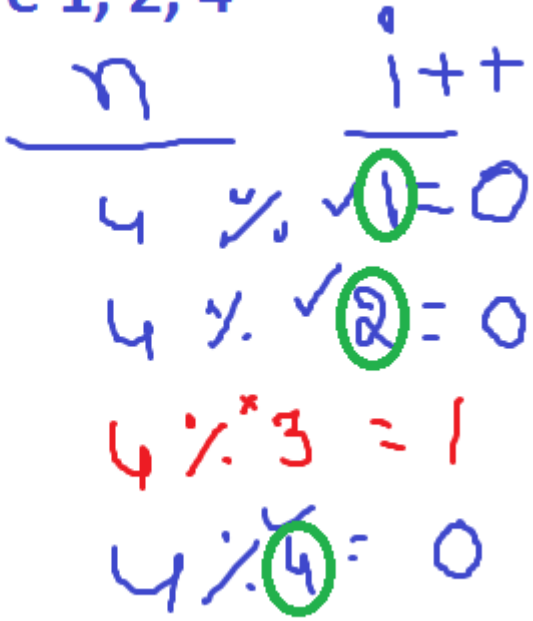

You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 14 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long int n; int s=0,r;
clrscr();
printf("Enter any no "); scanf("%ld",&n);
do
{
r=n%10; s=s+r;
n=n/10;
}while(n!=0);
printf("Digits sum=%d",s);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:20 PM
04-Jun-25

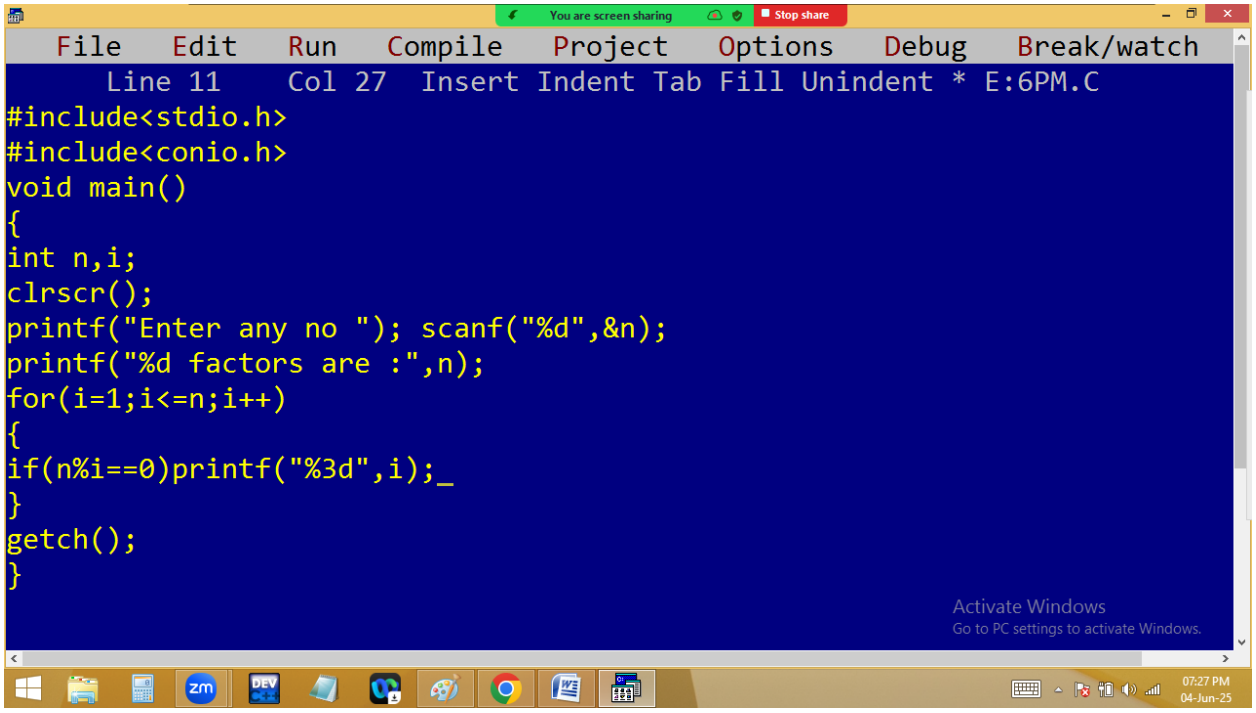

You are screen sharing
Stop share
Enter any no 456
Digits sum=15
Activate Windows
Go to PC settings to activate Windows.
07:20 PM
04-Jun-25

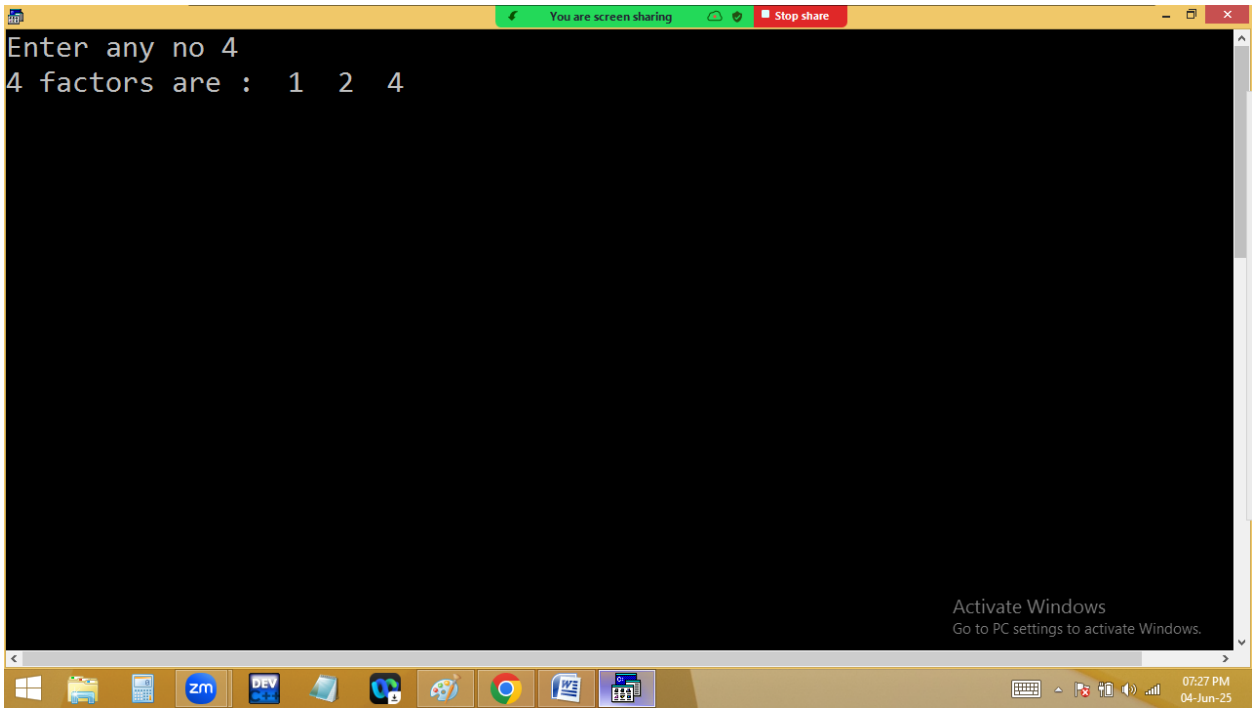
Printing factors of given no?

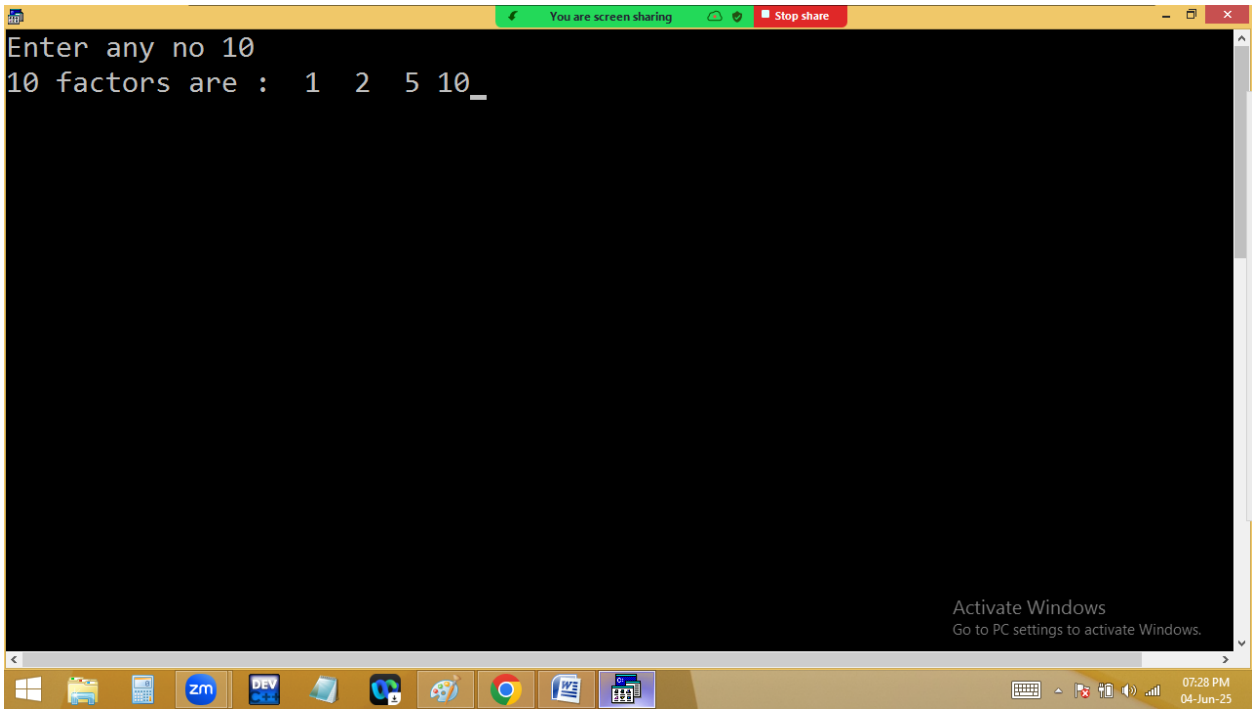
10 factors are 1, 2, 5, 10

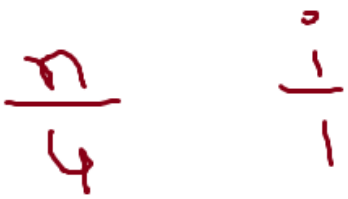
4 factors are 1, 2, 4




You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 27 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i;
clrscr();
printf("Enter any no "); scanf("%d",&n);
printf("%d factors are :",n);
for(i=1;i<=n;i++)
{
if(n%i==0)printf("%3d",i);_
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:27 PM
04-Jun-25


You are screen sharing
Stop share
Enter any no 4
4 factors are : 1 2 4
Activate Windows
Go to PC settings to activate Windows.
07:27 PM
04-Jun-25





```
for( i=1; i<=n; i++ )
{
if( n%i==0 ) p( i );
}
```

4%1=0==> 1

4%2=0==> 2

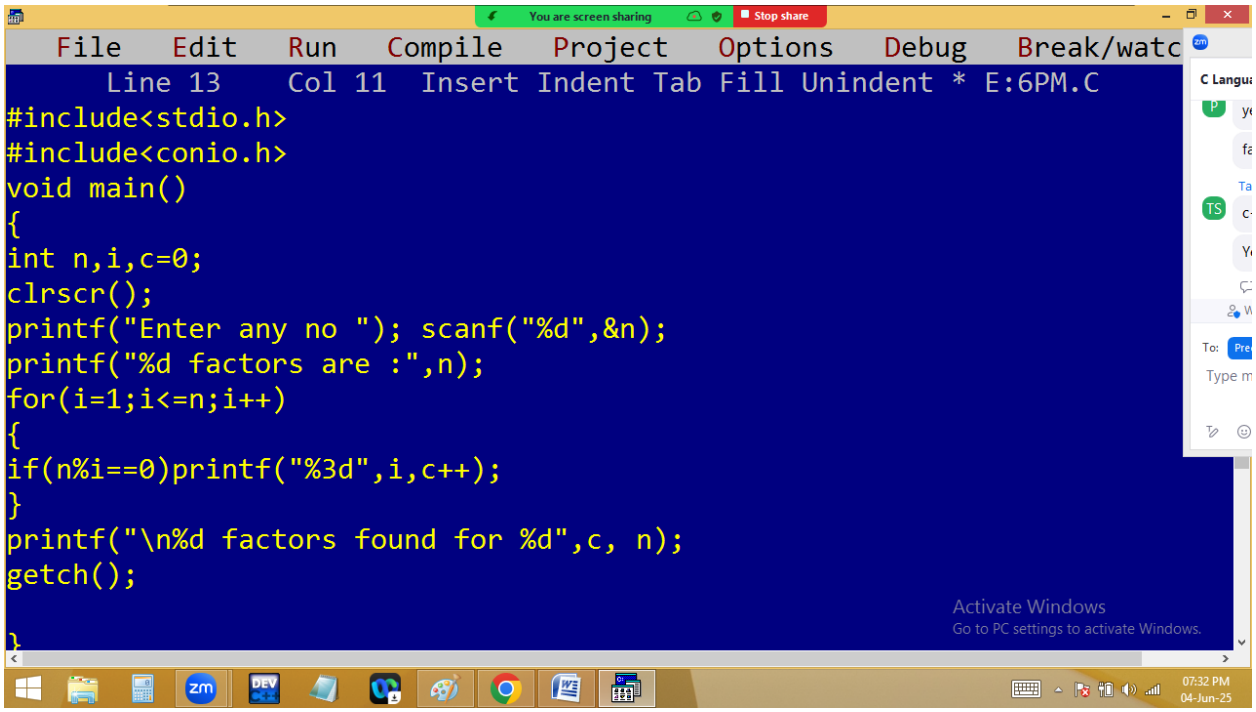
4%3=1==>Not

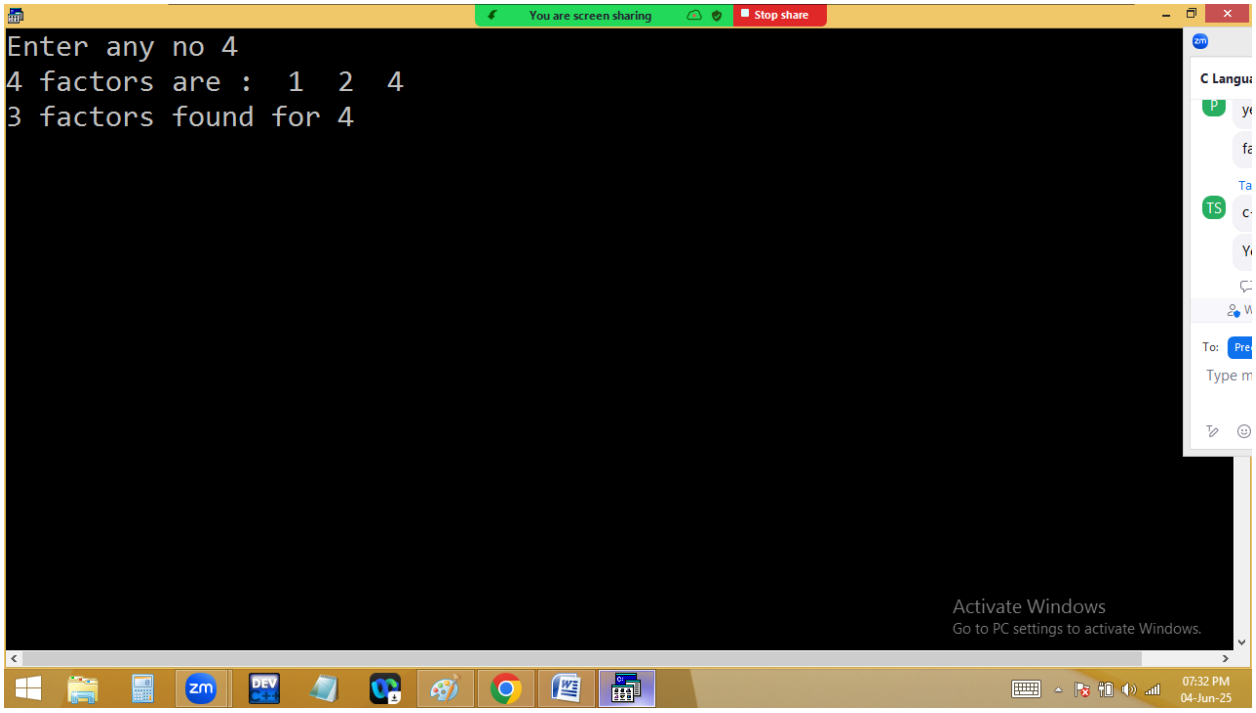
4%4=0==>4

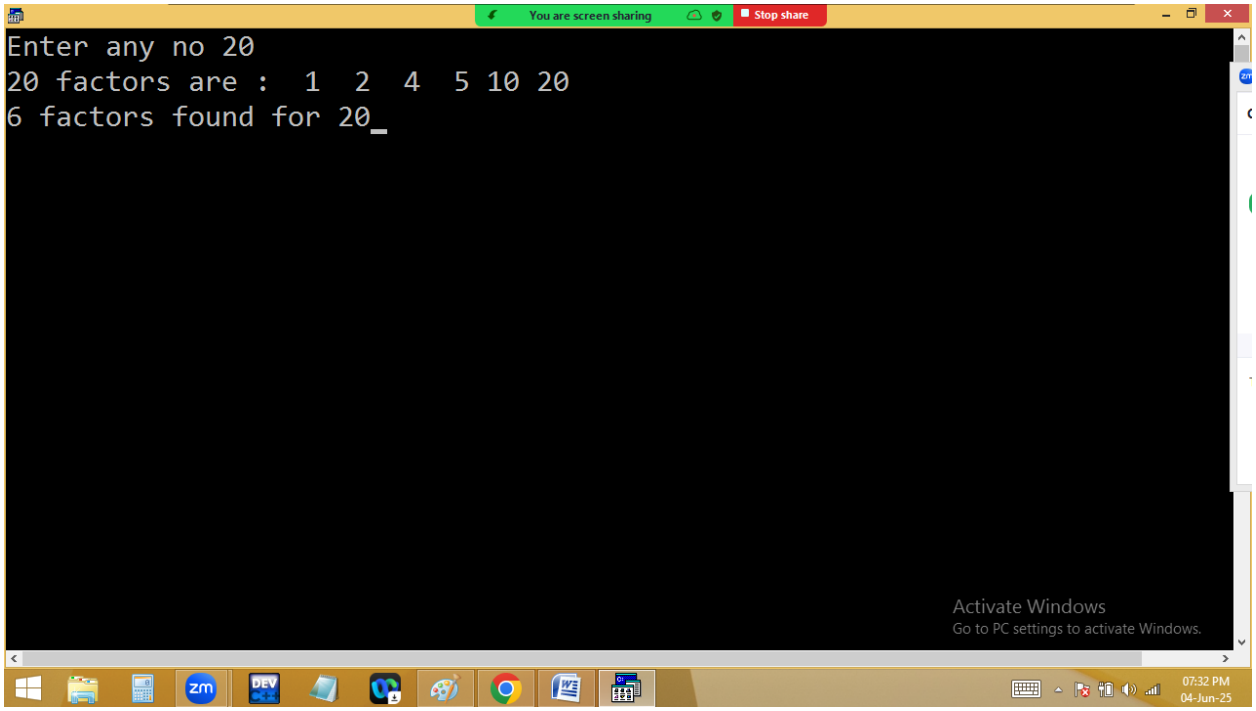
Printing factors of given no and count?

10 factors are 1 2 5 10

4 factors available




You are screen sharing
Stop share
Enter any no 4
4 factors are : 1 2 4
3 factors found for 4
Activate Windows
Go to PC settings to activate Windows.
07:32 PM
04-Jun-25



n / 4 | i / 1 | c / 0

```
for( i=1; i<=n; i++ )
{
if( n%i==0 ) p( i , c++ );
}
```

4%1=0==> 1

4%2=0==> 2

4%3=1==>Not

4%4=0==>4

p (c) ==> 3

Home work?

1. 5 ➔ 1+2+3+4+5=15

2. $5 \rightarrow 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$
3. $5 \rightarrow 1+1/1+1/2+1/3+1/4+1/5=3.28$